

Chapter 1

Probabilistic foundations

Throughout the natural sciences and our daily lives, we encounter a rich variety of phenomena involving some form stochasticity or randomness. The mathematical theory of probability provides a powerful way of understanding, interpreting and making predictions about such random phenomena.

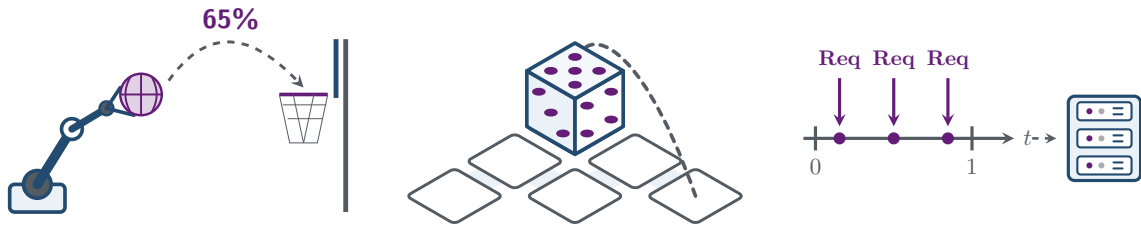


Figure 1.1. Examples of random phenomena (discussed in [Section 1.2.1](#)).

We begin with a brief review of sets and functions.

1.1 A brief summary of sets, functions summation

The purpose of this section is to review some background concepts on sets, functions and summation over potentially infinite sets. This will set the scene for the following discussion of discrete probability.

1.1.1 A few concepts from set theory

Here we give a brief review of set theoretic notation. For a more detailed introduction we recommend [BS11, Chapter 1]. Recall that a set A is an unordered collection of objects. We write $x \in A$ to mean that the object x is an element of the set A , and also write $x \notin A$ to mean that x is not an element of A . We say that A is a subset of B , denoted $A \subseteq B$, if whenever $x \in A$ we also have $x \in B$. The *empty set* is a set with no elements, denoted by $\emptyset$, so $x \in \emptyset$ is never true. Given a pair of sets A and B we let $A \setminus B$ denote the set of objects such that $x \in A$ but $x \notin B$. In particular, we always have $A = A \setminus \emptyset$. Given a set A and some property P , we let $\{x \in A : x \text{ has property } P\}$ denote the set of all elements $x \in A$ which have the property P .

The *power set* $\mathcal{P}(\Omega)$ of a set Ω denotes the set of all subsets of Ω , so $S \in \mathcal{P}(\Omega)$ if and only if $S \subseteq \Omega$. Suppose we have a collection $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ consisting of subsets $A \subseteq \Omega$. We define the *intersection* $\bigcap \mathcal{A}$ and *union* $\bigcup \mathcal{A}$ by

$$\begin{aligned}\bigcap \mathcal{A} &:= \{\omega \in \Omega : \omega \in A \text{ for every } A \in \mathcal{A}\} \\ \bigcup \mathcal{A} &:= \{\omega \in \Omega : \omega \in A \text{ for some } A \in \mathcal{A}\}.\end{aligned}$$

Given a pair of sets $A \subseteq \Omega$ and $B \subseteq \Omega$ we define their intersection $A \cap B := \bigcap \{A, B\}$, their union $A \cup B := \bigcup \{A, B\}$; and also let $A \setminus B$ denote the set of all elements $\omega \in A$ such that $\omega \notin B$. In the case of a sequence $(A_\ell)_{\ell \in \mathbb{I}}$ where $\mathbb{I} \subseteq \mathbb{N}$ is an initial segment, we use the standard notation

$$\begin{aligned}\bigcap_{\ell \in \mathbb{I}} A_\ell &:= \bigcap \{A_\ell : \ell \in \mathbb{I}\} = \{\omega \in \Omega : \omega \in A_\ell \text{ for every } \ell \in \mathbb{I}\}, \\ \bigcup_{\ell \in \mathbb{I}} A_\ell &:= \bigcup \{A_\ell : \ell \in \mathbb{I}\} = \{\omega \in \Omega : \omega \in A_\ell \text{ for some } \ell \in \mathbb{I}\}.\end{aligned}$$

Note that given a collection $\mathcal{A} \subseteq \mathcal{P}(\Omega)$, the intersection and union are connected through the relation $\bigcap \mathcal{A} = \Omega \setminus \bigcup \{\Omega \setminus A : A \in \mathcal{A}\}$.

Suppose we have some finite and non-empty index set $\mathbb{F}$ and for each $i \in \mathbb{F}$ we have a corresponding set A_i . We then define the *Cartesian product* by

$$\prod_{i \in \mathbb{F}} A_i := \left\{ g : \mathbb{F} \longrightarrow \bigcup_{i \in \mathbb{F}} A_i : g(i) \in A_i \text{ for every } i \in \mathbb{F} \right\}. \quad (1.1)$$

We write $A_1 \times \dots \times A_n$ for the special case where $\mathbb{F} = [n] = \{1, \dots, n\}$, often referred to as the n -fold Cartesian product. Given $a \in \prod_{i \in \mathbb{F}} A_i$ we often write $a_j = a(j)$ for $j \in \mathbb{F}$.

1.1.2 Functions and cardinality

Recall that a function $f : A \rightarrow B$ associates every element $a \in A$ with a unique element $f(a) \in B$. We let B^A denote the set of all functions $f : A \rightarrow B$ from A to B . Given $f \in B^A$, the set A is referred to as the *domain* of f and the set $\text{Img}(f) := \{f(a) : a \in A\} \subseteq B$ is referred to as the *image*. A function $f : A \rightarrow B$ is *surjective* if $\text{Img}(f) = B$. A function $f : A \rightarrow B$ is *injective* if $f(a_0) = f(a_1)$ implies $a_0 = a_1$. A function that is both injective and surjective is said *bijective*. We refer to a function which is bijective (respectively, injective, surjective) as a bijection (respectively, injection, surjection).

We let $\emptyset$ denote the empty set, let $\mathbb{Z}$ denote the integers, let $\mathbb{N} := \{\ell \in \mathbb{Z} : \ell > 0\}$ and $\mathbb{N}_0 := \{\ell \in \mathbb{Z} : \ell \geq 0\} = \mathbb{N} \cup \{0\}$. A set $\mathbb{F}$ is said to be *finite* if for some $n \in \mathbb{N}_0$ there exists a bijection $\varphi : [n] \rightarrow \mathbb{F}$ where

$$[n] := \{\ell \in \mathbb{N} : \ell \leq n\} = \begin{cases} \{1, \dots, n\} & \text{if } n \in \mathbb{N} \\ \emptyset & \text{if } n = 0. \end{cases}$$

When $\mathbb{F}$ is finite the number $n \in \mathbb{N}_0$ for which there is a bijection $\varphi : [n] \rightarrow \mathbb{F}$ is unique, and is referred to as the *cardinality* of $\mathbb{F}$ and denoted by $|\mathbb{F}|$. A set $\mathbb{A}$ is said to be *countably infinite* if there is a bijection $\varphi : \mathbb{N} \rightarrow \mathbb{A}$. Further more, a set $\mathbb{A}$ is said to be *countable* if $\mathbb{A}$ is either finite or countably infinite. We refer to $\mathbb{I} \subseteq \mathbb{N}$ as an *initial segment* if either $\mathbb{I} = \mathbb{N}$ or $\mathbb{I} = [n]$ for some $n \in \mathbb{N}_0$. Hence, a set $\mathbb{A}$ is countable if and only if there is a bijection $\varphi : \mathbb{I} \rightarrow \mathbb{A}$ for some initial segment $\mathbb{I} \subseteq \mathbb{N}$.

Suppose we have a function $f : A \rightarrow B$ so $f \in B^A$. Then, given $J \subseteq A$ we define a function $f|_J \in B^J$, with domain J , by $f|_J(\omega) = f(\omega)$ for all $\omega \in J$. We refer to $f|_J : J \rightarrow B$ as the *restriction* of f to J . We also let $f(J) := \text{Img}(f|_J) = \{f(\omega) : \omega \in J\} \subseteq B$, referred to as the *image* of J under f . In addition, given a function $f : A \rightarrow B$ and $K \subseteq B$ the *preimage* of K given by

$$f^{-1}(K) := \{\omega \in A : f(\omega) \in K\}. \quad (1.2)$$

Note that the preimage $f^{-1}(K)$ is well-defined even if f is not invertible.

1.1.3 Real valued functions

We let $\mathbb{R}$ denote the complete ordered field of *real numbers* [BS11, Chapter 2], with its usual algebraic operations $(+, \times, -)$ and total order $(\leq)$. We also let $\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, \infty\}$ denote the *extended real numbers* [Fol99, Section 0.5] and define $\text{lgt} : (0, 1) \rightarrow \mathbb{R}$ by $\text{lgt}(p) := \log\{p/(1-p)\}$ for all $p \in (0, 1)$. We also extend lgt to $\overline{\text{lgt}} : [0, 1] \rightarrow \overline{\mathbb{R}}$ by letting

$$\overline{\text{lgt}}(p) := \begin{cases} -\infty & \text{if } p = 0 \\ \text{lgt}(p) & \text{if } p \in (0, 1) \\ \infty & \text{if } p = 1. \end{cases}$$

Observe that lgt has strictly positive derivative $\text{lgt}'(p) = 1/\{p(1-p)\} > 0$ for all $p \in (0, 1)$ and $\text{lgt}\{(0, 1)\} = \mathbb{R}$. Indeed, lgt is an increasing bijection with inverse $\text{sgm} : \mathbb{R} \rightarrow (0, 1)$ given by $\text{sgm}(x) := 1/(1 + e^{-x})$. Consequently, the function $\overline{\text{lgt}} : [0, 1] \rightarrow \overline{\mathbb{R}}$ has inverse $\overline{\text{sgm}} : \overline{\mathbb{R}} \rightarrow [0, 1]$. Hence, we may extend the total order to $\overline{\mathbb{R}}$ by $\overline{\text{lgt}}(p) \leq \overline{\text{lgt}}(q)$ if and only if $p \leq q$ for all $p, q \in [0, 1]$. Hence, we have a total order on $\overline{\mathbb{R}}$ which extends the usual total order on $\mathbb{R}$ and satisfies

$$-\infty \leq a \leq \infty,$$

for all $a \in \overline{\mathbb{R}}$. In addition, the maps lgt and sgm are continuous, as are the binary operations $+$, $-$ and $\times$, considered as maps from $\mathbb{R}^2$ to $\mathbb{R}$. Hence, we may give a partial extension of $\star \in \{+, -, \times\}$ to certain triples $(a, b, c) \in \overline{\mathbb{R}}^3$ by $a \star b = c$ if and only if

$$\lim_{n \rightarrow \infty} \overline{\text{sgm}}\{\text{lgt}(p_n) \star \text{lgt}(q_n)\} = \overline{\text{sgm}}(c) \in [0, 1], \quad (1.3)$$

for every pair of sequences $(p_n)_{n=1}^\infty$ and $(q_n)_{n=1}^\infty$ in $(0, 1)$ such that $\lim_{n \rightarrow \infty} p_n = \overline{\text{sgm}}(a)$ and $\lim_{n \rightarrow \infty} q_n = \overline{\text{sgm}}(b)$. In particular, given $a \in \mathbb{R}$, we have

$$\begin{aligned} a + \infty &= \infty + a = \infty + \infty = \infty, \\ a + (-\infty) &= (-\infty) + a = (-\infty) + (-\infty) = (-\infty). \end{aligned}$$

By contrast, the sum $\infty + (-\infty)$ is not defined. Similarly, we extend the absolute value to $\overline{\mathbb{R}}$ by letting $|\infty| = |-\infty| = \infty$.

Given a set $A \subseteq \overline{\mathbb{R}}$ we say that $b \in \overline{\mathbb{R}}$ is an *upper bound* on the subset A if $a \leq b$ for all $a \in A$. We say that $s \in \overline{\mathbb{R}}$ is a *supremum*, denoted $s = \sup A$, if s is a least upper bound on A , so s is an upper bound on A and if $b \in \overline{\mathbb{R}}$ is an upper bound on A we have $s \leq b$. We say that $A \subseteq \overline{\mathbb{R}}$ is *bounded above* if there is a real number $b \in \mathbb{R}$ which is an upper bound on A . In particular, we have $\sup \emptyset = -\infty$ and if $\sup B = \infty$ if B has no upper bound in $\mathbb{R}$. A fundamental property of the real numbers is that, for every non-empty subset $A \subseteq \mathbb{R}$ which is bounded above has a supremum $\sup A \in \mathbb{R}$ [BS11]. Given a set $A \subseteq \overline{\mathbb{R}}$ and $\lambda \in \mathbb{R}$, such that either $\lambda \neq 0$ or $A \subseteq \mathbb{R}$, we let $\lambda A := \{\lambda \times a : a \in A\} \subseteq \overline{\mathbb{R}}$, and let $-A := (-1)A$. The *infimum* is defined by $\inf A := -\sup(-A)$. Given $a, b \in \overline{\mathbb{R}}$ we let $[a, b] := \{x \in \overline{\mathbb{R}} : a \leq x \leq b\}$, let $(a, b] := [a, b] \setminus \{a\}$, let $[a, b) := [a, b] \setminus \{b\}$, and let $(a, b) := [a, b] \setminus \{a, b\}$.

1.1.4 Summation over finite domains

Suppose we have a function $f : \Omega \rightarrow \overline{\mathbb{R}}$. Let's first suppose that $\mathbb{F} \subseteq \Omega$ is a finite set, of cardinality $n = |\mathbb{F}| \in \mathbb{N}_0$ such that $f(\mathbb{F}) = \text{Img}(f|_{\mathbb{F}}) = \{f(\omega) : \omega \in \mathbb{F}\} \subseteq (-\infty, \infty]$. Then, we have the familiar concept of summation given by

$$\sum_{\omega \in \mathbb{F}} f(\omega) = \sum_{\ell \in [n]} f\{\varphi(\ell)\} = \begin{cases} f\{\varphi(1)\} + \dots + f\{\varphi(n)\} & \text{if } n \in \mathbb{N} \\ 0 & \text{if } n = 0, \end{cases} \quad (1.4)$$

where $\varphi : [n] \rightarrow \mathbb{F}$ is a bijection.

Remark 1.1. The definition of $\sum_{\omega \in \mathbb{F}} f(\omega)$ given in (1.4) does not depend on the choice of bijection $\varphi : [n] \rightarrow \mathbb{F}$; a consequence of the fact that addition is *commutative* ($a + b = b + a$ for all $a, b \in \mathbb{R}$) and *associative* ($a + (b + c) = (a + b) + c$ for all $a, b, c \in \mathbb{R}$), and $a + \infty = \infty$ if $a \in \overline{\mathbb{R}} \setminus \{-\infty\}$.

Remark 1.2. We have assumed that $f(\mathbb{F}) = \text{Img}(f|_{\mathbb{F}}) \subseteq (-\infty, \infty]$. This ensures that the sum (1.4) is always well defined. Either $\text{Img}(f|_{\mathbb{F}}) \subseteq \mathbb{R}$, in which case $\sum_{\omega \in \mathbb{F}} f(\omega) \in \mathbb{R}$, or $\infty \in \text{Img}(f|_{\mathbb{F}})$, in which case $\sum_{\omega \in \mathbb{F}} f(\omega) = \infty$.

1.1.5 Summation over infinite domains

Now let's extend the concept of summation to potentially infinite sets.

Definition 1.3. Given a function $f : \Omega \rightarrow \overline{\mathbb{R}}$ on a set Ω we define

$$\|f\|_1 := \sup \left\{ \sum_{\omega \in \mathbb{F}} |f(\omega)| : \mathbb{F} \subseteq \Omega \text{ is a finite subset of } \Omega \right\} \in [0, \infty].$$

We then define $\ell^1(\Omega)$ and $\ell_+^1(\Omega) \subseteq \ell^1(\Omega)$ by

$$\begin{aligned} \ell^1(\Omega) &:= \{f : \Omega \rightarrow \overline{\mathbb{R}} : \|f\|_1 < \infty\} \\ \ell_+^1(\Omega) &:= \{f \in \ell^1(\Omega) : \text{Img}(f) \subseteq [0, \infty]\}. \end{aligned}$$

Observation 1.4. Given a function $f : \Omega \rightarrow \overline{\mathbb{R}}$ and any $t < \|f\|_1$ there exists a finite subset $\mathbb{F} \subseteq \Omega$ such that $t < \sum_{\omega \in \mathbb{F}} |f(\omega)| \leq \|f\|_1$.

Lemma 1.5. Suppose $f \in \ell^1(\Omega)$. Then $\text{Img}(f) \subseteq \mathbb{R}$.

Proof. Let's suppose that $\text{Img}(f) \subseteq \mathbb{R}$ is false and establish a contradiction. If $\text{Img}(f) \subseteq \mathbb{R}$ does not hold, then for some $\omega_\infty \in \Omega$ we have $f(\omega_\infty) \in \{-\infty, \infty\}$, so $|f(\omega_\infty)| = \infty$, and

hence $\|f\|_1 \geq \sum_{\omega \in \{\omega_\infty\}} |f(\omega)| = \infty$ by [Definition 1.3](#). This contradicts the assumption that $f \in \ell^1(\Omega)$, so $\|f\|_1 < \infty$. $\square$

Given a function $f : \Omega \rightarrow \overline{\mathbb{R}}$ we define $|f| : \Omega \rightarrow (-\infty, \infty]$ by $|f|(\omega) = |f(\omega)|$ for all $\omega \in \Omega$.

Lemma 1.6. Suppose we have a function $f : \Omega \rightarrow \overline{\mathbb{R}}$. Then $\|f\|_1 = \||f|\|_1 \geq 0$. In particular, we have $f \in \ell^1(\Omega)$ if and only if $|f| \in \ell^1_+(\Omega)$.

Proof of Lemma 1.6. Given a function $f : \Omega \rightarrow \overline{\mathbb{R}}$ we let

$$\text{FS}_+(f) := \left\{ \sum_{\omega \in \mathbb{F}} |f(\omega)| : \mathbb{F} \subseteq \Omega \text{ is a finite subset of } \Omega \right\} \subseteq [0, \infty], \quad (1.5)$$

so that $\|f\|_1 = \sup \text{FS}_+(f)$. Since $\emptyset \subseteq \Omega$ is a finite set we have $0 = \sum_{\omega \in \emptyset} |f(\omega)| \in \text{FS}_+(f)$, so that $\|f\|_1 \geq 0$. Furthermore, we also have $\text{FS}_+(|f|) = \text{FS}_+(f)$ so that $\||f|\|_1 = \|f\|_1 \geq 0$. Since $\text{Img}(|f|) \subseteq [0, \infty]$ so that $f \in \ell^1(\Omega)$ if and only if $\||f|\|_1 = \|f\|_1 \in [0, \infty)$ so if and only if $|f| \in \ell^1_+(\Omega)$. $\square$

Given a function $f : \Omega \rightarrow \overline{\mathbb{R}}$ the *cozero set* of f is defined by

$$\text{coz}(f) := \{ \omega \in \Omega : f(\omega) \neq 0 \} = f^{-1}(\overline{\mathbb{R}} \setminus \{0\}). \quad (1.6)$$

Lemma 1.7. Suppose $f : \Omega \rightarrow \overline{\mathbb{R}}$ is a function. If $f \in \ell^1(\Omega)$ the set $\text{coz}(f)$ is countable, and also $\text{Img}(f) \subseteq \mathbb{R}$. Similarly, if $f \in \ell^1_+(\Omega)$ then $\text{Img}(f) \subseteq [0, \infty)$.

Proof of Lemma 1.7. Given any $n \in \mathbb{N}$ let

$$A_n := \{ \omega \in \Omega : |f(\omega)| > 1/n \} = f^{-1}(\overline{\mathbb{R}} \setminus [-1/n, 1/n]).$$

Observe that for every $\omega \in \text{coz}(f)$ we have $|f(\omega)| \in (0, \infty)$ ([Lemma 1.5](#)), so there exists some $n \in \mathbb{N}$ such that $|f(\omega)| > 1/n$ [[BS11](#), Corollary 2.4.5] and $\omega \in A_n$. Thus, we have $\text{coz}(f) = \bigcup \{A_n : n \in \mathbb{N}\}$.

Since $f \in \ell^1(\Omega)$ so $\|f\|_1 < \infty$, so $m > n \|f\|_1$ [[BS11](#), Archimedean Property 2.4.3] for some $m \in \mathbb{N}$. If A_n is not finite, for some $n \in \mathbb{N}$, then there exists a finite subset $\mathbb{F}_n \subseteq A_n$ with cardinality $|\mathbb{F}_n| = m > n \|f\|_1$, so

$$\|f\|_1 \geq \sum_{\omega \in \mathbb{F}_n} |f(\omega)| \geq \frac{|\mathbb{F}_n|}{n} > \|f\|_1,$$

which yields a contradiction. Hence, each set A_n must be finite. Consequently, the set $\text{coz}(f) = \bigcup \{A_n : n \in \mathbb{N}\}$ is a countable union of finite sets, so must also be countable [[BS11](#), Theorem 1.3.12]. $\square$

Given functions $f : \Omega \rightarrow \mathbb{R}$, $g : \Omega \rightarrow \mathbb{R}$ and $\lambda \in \mathbb{R}$ we define functions $(f + g) : \Omega \rightarrow \mathbb{R}$, $(fg) : \Omega \rightarrow \mathbb{R}$, $(\lambda f) : \Omega \rightarrow \mathbb{R}$ and $|f| : \Omega \rightarrow \mathbb{R}$ by $(f + g)(\omega) = f(\omega) + g(\omega)$, $(fg)(\omega) = f(\omega)g(\omega)$, $(\lambda f)(\omega) = \lambda f(\omega)$ and $|f|(\omega) = |f(\omega)|$ for all $\omega \in \Omega$.

Lemma 1.8. Suppose that $f : \Omega \rightarrow \mathbb{R}$, $g : \Omega \rightarrow \mathbb{R}$. Then, we have $\|f + g\|_1 \leq \|f\|_1 + \|g\|_1$. Hence, if $f, g \in \ell^1(\Omega)$ then $f + g \in \ell^1(\Omega)$. Furthermore, given $f \in \ell^1(\Omega)$ and $\lambda \in \mathbb{R}$ we have $\|\lambda f\|_1 = |\lambda| \|f\|_1$ and $\lambda f \in \ell^1(\Omega)$.

Proof of Lemma 1.8. We define $\text{FS}_+(h)$ as in (1.5) for functions $h : \Omega \rightarrow \mathbb{R}$. If $\mathbb{F} \subseteq \Omega$ is a finite set then

$$\sum_{\omega \in \mathbb{F}} |f + g|(\omega) \leq \sum_{\omega \in \mathbb{F}} |f|(\omega) + \sum_{\omega \in \mathbb{F}} |g|(\omega) \leq \sup \text{FS}_+(|f|) + \sup \text{FS}_+(|g|) = \|f\|_1 + \|g\|_1.$$

Taking a supremum over finite subsets $\mathbb{F} \subseteq \Omega$ yields $\|f + g\|_1 \leq \|f\|_1 + \|g\|_1$. Thus, if f and $g \in \ell^1(\Omega)$ then $f + g \in \ell^1(\Omega)$.

Suppose that $f \in \ell^1(\Omega)$ and $\lambda \in \mathbb{R}$. Given any finite set $\mathbb{F} \subseteq \Omega$ we have $\sum_{\omega \in \mathbb{F}} |\lambda f|(\omega) = |\lambda| (\sum_{\omega \in \mathbb{F}} |f|(\omega))$, so that $\text{FS}_+(|\lambda f|) = |\lambda| \text{FS}_+(|f|)$ and consequently $\|\lambda f\|_1 = |\lambda| \|f\|_1$. Thus $\lambda f \in \ell^1(\Omega)$ since $f \in \ell^1(\Omega)$. $\square$

Given a subset $\mathbb{S} \subseteq \Omega$ we define $\mathbf{1}_{\mathbb{S}} : \Omega \rightarrow \mathbb{R}$

$$\mathbf{1}_{\mathbb{S}}(\omega) := \begin{cases} 0 & \text{if } \omega \notin \mathbb{S} \\ 1 & \text{if } \omega \in \mathbb{S}. \end{cases}$$

Hence, given a function $f : \Omega \rightarrow \mathbb{R}$ the function $(f \mathbf{1}_{\mathbb{S}}) : \Omega \rightarrow \mathbb{R}$ satisfies $(f \mathbf{1}_{\mathbb{S}})(\omega) = f(\omega)$ when $\omega \in \mathbb{S}$ and $(f \mathbf{1}_{\mathbb{S}})(\omega) = 0$ when $\omega \in \Omega \setminus \mathbb{S}$.

Lemma 1.9. Suppose that $f : \Omega \rightarrow \mathbb{R}$ and $\mathbb{S} \subseteq \Omega$. Then $\|f\|_1 = \|f \mathbf{1}_{\mathbb{S}}\|_1 + \|f \mathbf{1}_{\Omega \setminus \mathbb{S}}\|_1$.

Proof of Lemma 1.9. Observe that since $f = f \mathbf{1}_{\mathbb{S}} + f \mathbf{1}_{\Omega \setminus \mathbb{S}}$ by Lemma 1.8 we have

$$\|f\|_1 = \|f \mathbf{1}_{\mathbb{S}} + f \mathbf{1}_{\Omega \setminus \mathbb{S}}\|_1 \leq \|f \mathbf{1}_{\mathbb{S}}\|_1 + \|f \mathbf{1}_{\Omega \setminus \mathbb{S}}\|_1.$$

Next, take $s < \|f \mathbf{1}_{\mathbb{S}}\|_1$ and $t < \|f \mathbf{1}_{\Omega \setminus \mathbb{S}}\|_1$. By Observation 1.4 there exist finite subsets $\mathbb{F} \subseteq \Omega$ and $\mathbb{G} \subseteq \Omega$ such that $\sum_{\omega \in \mathbb{F}} |(f \mathbf{1}_{\mathbb{S}})(\omega)| > s$ and $\sum_{\omega \in \mathbb{G}} |(f \mathbf{1}_{\Omega \setminus \mathbb{S}})(\omega)| > t$. Hence, the finite subset $\mathbb{F} \cup \mathbb{G} \subseteq \Omega$ we have

$$\begin{aligned} s + t &< \sum_{\omega \in \mathbb{F}} |(f \mathbf{1}_{\mathbb{S}})(\omega)| + \sum_{\omega \in \mathbb{G}} |(f \mathbf{1}_{\Omega \setminus \mathbb{S}})(\omega)| \\ &\leq \sum_{\omega \in \mathbb{F} \cup \mathbb{G}} |(f \mathbf{1}_{\mathbb{S}})(\omega)| + \sum_{\omega \in \mathbb{F} \cup \mathbb{G}} |(f \mathbf{1}_{\Omega \setminus \mathbb{S}})(\omega)| \\ &= \sum_{\omega \in (\mathbb{F} \cup \mathbb{G}) \cap \mathbb{S}} |f(\omega)| + \sum_{\omega \in (\mathbb{F} \cup \mathbb{G}) \setminus \mathbb{S}} |f(\omega)| \\ &= \sum_{\omega \in \mathbb{F} \cup \mathbb{G}} |f(\omega)| \leq \|f\|_1 \leq \|f \mathbf{1}_{\mathbb{S}}\|_1 + \|f \mathbf{1}_{\Omega \setminus \mathbb{S}}\|_1. \end{aligned}$$

Letting $s \nearrow \|f \mathbf{1}_{\mathbb{S}}\|_1$ and $t \nearrow \|f \mathbf{1}_{\Omega \setminus \mathbb{S}}\|_1$ we deduce that $\|f\|_1 = \|f \mathbf{1}_{\mathbb{S}}\|_1 + \|f \mathbf{1}_{\Omega \setminus \mathbb{S}}\|_1$. $\square$

Recall from Section 1.1.2 that given $f : \Omega \rightarrow \overline{\mathbb{R}}$ and $\mathbb{S} \subseteq \Omega$, the restriction $f|_{\mathbb{S}} : \mathbb{S} \rightarrow \overline{\mathbb{R}}$ is defined by $f|_{\mathbb{S}}(\omega) = f(\omega)$ for all $\omega \in \mathbb{S}$.

Lemma 1.10. Suppose that $f : \Omega \rightarrow \mathbb{R}$ and $\mathbb{S} \subseteq \Omega$. Then $\|f|_{\mathbb{S}}\|_1 = \|f\mathbf{1}_{\mathbb{S}}\|_1$. Hence, we have $\|f\|_1 = \|f\mathbf{1}_{\mathbb{S}}\|_1 + \|f\mathbf{1}_{\Omega \setminus \mathbb{S}}\|_1 = \|f|_{\mathbb{S}}\|_1 + \|f|_{\Omega \setminus \mathbb{S}}\|_1$.

Proof of Lemma 1.10. Given any $\mathbb{F} \subseteq \mathbb{S}$ we have

$$\sum_{\omega \in \mathbb{F}} |f|_{\mathbb{S}}(\omega) = \sum_{\omega \in \mathbb{F}} |f\mathbf{1}_{\mathbb{S}}(\omega)| \leq \|f\mathbf{1}_{\mathbb{S}}\|_1,$$

so $\|f|_{\mathbb{S}}\|_1 \leq \|f\mathbf{1}_{\mathbb{S}}\|_1$. On the other hand, given any finite set $\mathbb{G} \subseteq \Omega$

$$\begin{aligned} \sum_{\omega \in \mathbb{G}} |f\mathbf{1}_{\mathbb{S}}(\omega)| &= \sum_{\omega \in \mathbb{G} \cap \mathbb{S}} |f\mathbf{1}_{\mathbb{S}}(\omega)| + \sum_{\omega \in \mathbb{G} \setminus \mathbb{S}} |f\mathbf{1}_{\mathbb{S}}(\omega)| \\ &= \sum_{\omega \in \mathbb{G} \cap \mathbb{S}} |f|_{\mathbb{S}}(\omega) + \sum_{\omega \in \mathbb{G} \setminus \mathbb{S}} 0 = \sum_{\omega \in \mathbb{G} \cap \mathbb{S}} |f|_{\mathbb{S}}(\omega) \leq \|f|_{\mathbb{S}}\|_1. \end{aligned}$$

Taking a supremum over all finite sets $\mathbb{G} \subseteq \Omega$ yields $\|f\mathbf{1}_{\mathbb{S}}\|_1 \leq \|f|_{\mathbb{S}}\|_1$. Combining with the previous inequality $\|f|_{\mathbb{S}}\|_1 \leq \|f\mathbf{1}_{\mathbb{S}}\|_1$ we have $\|f|_{\mathbb{S}}\|_1 = \|f\mathbf{1}_{\mathbb{S}}\|_1$.

The second part of the lemma follows from Lemma 1.9. $\square$

Lemma 1.11. Suppose that $f : \Omega \rightarrow [0, \infty]$, $g : \Omega \rightarrow [0, \infty]$ and $\lambda \in (0, \infty)$. Then, we have $\|\lambda f\|_1 = |\lambda| \|f\|_1$ and $\|f + g\|_1 = \|f\|_1 + \|g\|_1$. Hence, if $f, g \in \ell_+^1(\Omega)$ then $\lambda f \in \ell_+^1(\Omega)$ and $f + g \in \ell_+^1(\Omega)$.

Proof of Lemma 1.11. Let's first show that $\|\lambda f\|_1 = |\lambda| \|f\|_1$. If $\|f\|_1 = \infty$ then $\|\lambda f\|_1 = \infty = \lambda \times \infty = |\lambda| \|f\|_1$ and case where $\|f\|_1 < \infty$ follows from Lemma 1.8.

Next, we establish the claim $\|f + g\|_1 = \|f\|_1 + \|g\|_1$. If $\infty \in \text{Img}(f) \cup \text{Img}(g)$ then for some $\omega_0 \in \Omega$ we have $\infty \in \{f(\omega_0), g(\omega_0)\}$, so $f(\omega_0) + g(\omega_0) = \infty$ and hence $\|f + g\|_1 = \|f\|_1 + \|g\|_1 = \infty$. Now let's suppose $\infty \notin \text{Img}(f) \cup \text{Img}(g)$. By Lemma 1.8 we have $\|f + g\|_1 \leq \|f\|_1 + \|g\|_1$ and choose $s < \|f\|_1$ and $t < \|g\|_1$. By Observation 1.4 we may choose finite subsets $\mathbb{F} \subseteq \Omega$ and $\mathbb{G} \subseteq \Omega$ such that $s < \sum_{\omega \in \mathbb{F}} f(\omega) = \sum_{\omega \in \mathbb{F}} |f(\omega)|$ and $t < \sum_{\omega \in \mathbb{G}} g(\omega) = \sum_{\omega \in \mathbb{G}} |g(\omega)|$. Hence, we have

$$\begin{aligned} s + t &\leq \sum_{\omega \in \mathbb{F}} f(\omega) + \sum_{\omega \in \mathbb{G}} g(\omega) \leq \sum_{\omega \in \mathbb{F} \cup \mathbb{G}} f(\omega) + \sum_{\omega \in \mathbb{F} \cup \mathbb{G}} g(\omega) \\ &= \sum_{\omega \in \mathbb{F} \cup \mathbb{G}} (f + g)(\omega) \leq \|f + g\|_1 \leq \|f\|_1 + \|g\|_1, \end{aligned}$$

Taking $s \nearrow \|f\|_1$ and $t \nearrow \|g\|_1$ yields the result $\|f\|_1 + \|g\|_1 = \|f + g\|_1$. $\square$

Corollary 1.12. Suppose that $f : \Omega \rightarrow \overline{\mathbb{R}}$ and $h : \Omega \rightarrow \overline{\mathbb{R}}$ satisfy $0 \leq |f(\omega)| \leq |h(\omega)|$ for all $\omega \in \Omega$. Then, $\|f\|_1 \leq \|h\|_1$. In particular, if $h \in \ell^1(\Omega)$ then $f \in \ell^1(\Omega)$.

Proof of Corollary 1.12. To prove $\|f\|_1 \leq \|h\|_1$ we first assume that $f : \Omega \rightarrow [0, \infty]$ and $h : \Omega \rightarrow [0, \infty]$ are non-negative so that $0 \leq f(\omega) \leq h(\omega)$ for all $\omega \in \Omega$. If $\infty \in \text{Img}(h)$ then $\|f\|_1 \leq \infty = \|h\|_1$, so we may suppose, without loss of generality, that $\infty \notin \text{Img}(h)$.

Given $\infty \notin \text{Img}(h)$, we have $\infty \notin \text{Img}(f)$. Thus, we have $\text{Img}(f) \cup \text{Img}(h) \subseteq [0, \infty)$. Hence, we may define $(h - f) : \Omega \rightarrow [0, \infty)$ by $(h - f)(\omega) = h(\omega) - f(\omega)$ for all $\omega \in \Omega$. Hence, Lemma 1.11 yields

$$\|f\|_1 + \|h - f\|_1 = \|f + (h - f)\|_1 = \|h\|_1,$$

so by Lemma 1.6 we have $\|h - f\|_1 \geq 0$. This completes the proof $\|f\|_1 \leq \|h\|_1$ for non-negative functions $f : \Omega \rightarrow [0, \infty]$ and $h : \Omega \rightarrow [0, \infty]$.

More generally, given any pair of functions $f : \Omega \rightarrow \overline{\mathbb{R}}$ and $h : \Omega \rightarrow \overline{\mathbb{R}}$ satisfy $0 \leq |f(\omega)| \leq |h(\omega)|$ for all $\omega \in \Omega$ the bound $\|f\|_1 \leq \|h\|_1$ follows from the non-negative special case. By Lemma 1.6 we deduce that $\|f\|_1 \leq \|h\|_1$. $\square$

We say that $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is a collection of pairwise disjoint subsets if $A_0 \cap A_1 = \emptyset$ for any pair of sets $A_0 \in \mathcal{A}$ and $A_1 \in \mathcal{A}$ such that $A_0 \neq A_1$. Given $\mathbb{S} \subseteq \Omega$, we say that $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is a *partition* of $\mathbb{S}$ if $\mathcal{A}$ is a collection of pairwise disjoint subsets such that $\mathbb{S} = \bigcup \mathcal{A}$.

Lemma 1.13. Suppose $f : \Omega \rightarrow \overline{\mathbb{R}}$ is a function, and $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is a partition of Ω . Let $G : \mathcal{A} \rightarrow [0, \infty]$ denote the function given by $G(B) := \|f|_B\|_1$ for all $B \in \mathcal{A}$. Then $\|f\|_1 = \|G\|_1$.

Note that the special case of Lemma 1.13 in which Ω is finite from the commutativity and associativity of addition since for each element $\omega \in \bigcup \mathcal{A}$ there is a unique $B \in \mathcal{A}$ such that $\omega \in B$.

Proof of Lemma 1.13. First note that for each $B \in \mathcal{A}$, given any finite set $\mathbb{F} \subseteq B \subseteq \Omega$, so that

$$\sum_{\omega \in \mathbb{F}} |(f|_B)(\omega)| = \sum_{\omega \in \mathbb{F}} |f(\omega)| \leq \|f\|_1,$$

so $G(B) \leq \|f\|_1$. In particular, if $\infty \in \text{Img}(G)$, so $G(B) = \infty$ for some $B \in \mathcal{A}$ then $\|G\|_1 = \infty = \|f\|_1$. Henceforth, let's consider the case where $\text{Img}(G) \subseteq [0, \infty)$.

Suppose $\mathcal{A}_o \subseteq \mathcal{A}$ is a non-empty finite subset. Take $\varepsilon > 0$ and for each $B \in \mathcal{A}_o$ choose a finite subset $J(B) \subseteq B$ such that

$$\sum_{\omega \in J(B)} |f(\omega)| > \|f|_B\|_1 - \frac{\varepsilon}{|\mathcal{A}_o|} = G(B) - \frac{\varepsilon}{|\mathcal{A}_o|},$$

which must exist by Observation 1.4 since $G(B) \in [0, \infty)$. Given any $B_0, B_1 \in \mathcal{A}$ such that $B_0 \neq B_1$ we have $B_0 \cap B_1 = \emptyset$ and so $J(B_0) \cap J(B_1) = \emptyset$. Hence, $\{J(B) : B \in \mathcal{A}_o\}$ forms a collection of finitely many disjoint finite subsets of Ω , so $K := \bigcup \{J(B) : B \in \mathcal{A}_o\} \subseteq \Omega$ is a finite subset and

$$\begin{aligned} \|f\|_1 &\geq \sum_{\omega \in K} |f(\omega)| = \sum_{B \in \mathcal{A}_o} \sum_{\omega \in J(B)} |f(\omega)| \\ &> \sum_{B \in \mathcal{A}_o} \left(G(B) - \frac{\varepsilon}{|\mathcal{A}_o|} \right) = \sum_{B \in \mathcal{A}_o} |G(B)| - \varepsilon. \end{aligned}$$

Taking $\varepsilon \searrow 0$ we deduce that $\sum_{B \in \mathcal{A}_o} |G(B)| \leq \|f\|_1$. If $\mathcal{A}_o = \emptyset$ then we have $\sum_{B \in \mathcal{A}_o} |G(B)| = 0 \leq \|f\|_1$. Since $\sum_{B \in \mathcal{A}_o} |G(B)| \leq \|f\|_1$ holds for all finite subsets $\mathcal{A}_o \subseteq \mathcal{A}$ we deduce $\|G\|_1 \leq \|f\|_1$.

Next, let's show that $\|f\|_1 \leq \|G\|_1$. Suppose $\mathbb{F} \subseteq \Omega$ is a finite subset and let $\mathcal{A}(\mathbb{F}) := \{B : B \in \mathcal{A} \text{ and } B \cap \mathbb{F} \neq \emptyset\} \subseteq \mathcal{A}$. Since $\mathcal{A}$ is a partition of Ω , it follows that $\{B \cap \mathbb{F} : B \in \mathcal{A}(\mathbb{F})\}$ is partition of the finite set $\mathbb{F}$. Consequently, we have

$$\begin{aligned} \sum_{\omega \in \mathbb{F}} |f(\omega)| &= \sum_{B \in \mathcal{A}(\mathbb{F})} \left(\sum_{\omega \in B \cap \mathbb{F}} |f(\omega)| \right) \\ &\leq \sum_{B \in \mathcal{A}(\mathbb{F})} \|f|_B\|_1 = \sum_{B \in \mathcal{A}(\mathbb{F})} G(B) \leq \|G\|_1. \end{aligned}$$

Since this holds for all finite sets $\mathbb{F} \subseteq \Omega$ we have $\|f\|_1 \leq \|G\|_1$. $\square$

Given a function $f : \Omega \rightarrow \overline{\mathbb{R}}$ we define non-negative functions $f_+ : \Omega \rightarrow [0, \infty]$ and $f_- : \Omega \rightarrow [0, \infty]$ by

$$\begin{aligned} f_+(\omega) &:= \max\{f(\omega), 0\} \\ f_-(\omega) &:= \max\{-f(\omega), 0\}. \end{aligned} \tag{1.7}$$

Observe that $f(\omega) = f_+(\omega) - f_-(\omega)$ for all $\omega \in \Omega$.

Lemma 1.14. Given any function $f : \Omega \rightarrow \overline{\mathbb{R}}$ we have $\|f\|_1 = \|f_-\|_1 + \|f_+\|_1$. In particular, we have $f \in \ell^1(\Omega)$ if and only if both $f_- \in \ell^1_+(\Omega)$ and $f_+ \in \ell^1_+(\Omega)$.

Proof of Lemma 1.14. If $\{-\infty, \infty\} \cap \text{Img}(f) \neq \emptyset$ then $\|f\|_1 = \|f_-\|_1 + \|f_+\|_1 = \infty$. On the other hand, if $\text{Img}(f) \subseteq \mathbb{R}$ then by applying Lemma 1.9 with $\mathbb{S} = \{\omega \in \Omega : f(\omega) > 0\}$ yields $\|f\|_1 = \|f_-\|_1 + \|f_+\|_1$. $\square$

Definition 1.15. Suppose we have a function $f : \Omega \rightarrow \overline{\mathbb{R}}$ and $\mathbb{S} \subseteq \Omega$ such that $\min\{\|(f|_{\mathbb{S}})_+\|_1, \|(f|_{\mathbb{S}})_-\|_1\} < \infty$. We define

$$\sum_{\omega \in \mathbb{S}} f(\omega) := \|(f|_{\mathbb{S}})_+\|_1 - \|(f|_{\mathbb{S}})_-\|_1.$$

Note that constraint $\min\{\|(f|_{\mathbb{S}})_+\|_1, \|(f|_{\mathbb{S}})_-\|_1\} < \infty$ ensures that $\sum_{\omega \in \mathbb{S}} f(\omega)$ is well-defined by Definition 1.15. Moreover, by Lemma 1.14 if $f|_{\mathbb{S}} \in \ell^1(\mathbb{S})$ then both $\|(f|_{\mathbb{S}})_+\|_1, \|(f|_{\mathbb{S}})_-\|_1 \in [0, \infty)$ are finite. Furthermore, when $\mathbb{S}$ is itself a finite set then Definition 1.15 agrees with our familiar definition of summation (1.4), whenever both definitions apply. A key feature of the form of summation given by Definition 1.15 is that it does not depend on a particular choice of ordering of the set $\mathbb{S}$.

Caution 1.16. There are different approaches to extending the concept of summation beyond the finite realm. A popular alternative is to take limits of a truncated sum. For example, consider the function $f : \mathbb{N} \rightarrow \mathbb{R}$ by $f(n) := -(-1)^n/n$ for all $n \in \mathbb{N}$. Then, we have

$$\lim_{m \rightarrow \infty} \left\{ \sum_{n=1}^m f(n) \right\} = \lim_{m \rightarrow \infty} \left\{ f(1) + \dots + f(m) \right\} = \log(2).$$

However, given any real number $a \in \mathbb{R}$ there is a bijection $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$\lim_{m \rightarrow \infty} \left\{ \sum_{n=1}^m f(\varphi\{n\}) \right\} = \lim_{m \rightarrow \infty} \left\{ f(\varphi\{1\}) + \dots + f(\varphi\{m\}) \right\} = a.$$

This follows a result of Riemann on the rearrangement of series [BS11, Section 9.1] from since with the decomposition $f = f_+ - f_-$ (1.7) we have $\|f_+\|_1 = \|f_-\|_1 = \infty$. In contrast, our choice (Definition 1.15) has the advantage that it does not depend on a particular choice of enumeration of the set $\mathbb{S}$.

Lemma 1.17. Suppose $\mathbb{S} \subseteq \Omega$, $f \in \ell^1(\Omega)$ and $g \in \ell^1(\Omega)$ satisfy $f(\omega) \leq g(\omega)$ for all $\omega \in \mathbb{S}$. Then, we have

$$\sum_{\omega \in \mathbb{S}} f(\omega) \leq \sum_{\omega \in \mathbb{S}} g(\omega).$$

Proof of Lemma 1.17. Applying the decomposition (1.7) we see that for all $\omega \in \mathbb{S}$

$$f_+(\omega) - f_-(\omega) = f(\omega) \leq g(\omega) = g_+(\omega) - g_-(\omega),$$

so that $f_+(\omega) + g_-(\omega) \leq f_-(\omega) + g_+(\omega)$. Hence, by Corollary 1.12 and Lemma 1.11 we have

$$\|(f|_{\mathbb{S}})_+\|_1 + \|(g|_{\mathbb{S}})_-\|_1 \leq \|(f|_{\mathbb{S}})_-\|_1 + \|(g|_{\mathbb{S}})_+\|_1.$$

The conclusion of the lemma now follows by Definition 1.15. $\square$

Lemma 1.18. Suppose $\mathbb{S} \subseteq \Omega$, and $f, g : \Omega \rightarrow \mathbb{R}$ are functions with $f|_{\mathbb{S}}, g|_{\mathbb{S}} \in \ell^1(\mathbb{S})$. Then,

$$\sum_{\omega \in \mathbb{S}} (f + g)(\omega) = \left(\sum_{\omega \in \mathbb{S}} f(\omega) \right) + \left(\sum_{\omega \in \mathbb{S}} g(\omega) \right).$$

Further more, given $\alpha \in \mathbb{R}$ we have $\sum_{\omega \in \mathbb{S}} (\alpha f)(\omega) = \alpha \sum_{\omega \in \mathbb{S}} f(\omega)$.

Proof of Lemma 1.18. Given any $\omega \in \mathbb{S}$ we have

$$\begin{aligned} (\{f+g\}|_{\mathbb{S}})_+(\omega) - (\{f+g\}|_{\mathbb{S}})_-(\omega) &= (\{f+g\}|_{\mathbb{S}})(\omega) = f(\omega) + g(\omega) \\ &= (f|_{\mathbb{S}})(\omega) + (g|_{\mathbb{S}})(\omega) \\ &= (f|_{\mathbb{S}})_+(\omega) - (f|_{\mathbb{S}})_-(\omega) + (g|_{\mathbb{S}})_+(\omega) - (g|_{\mathbb{S}})_-(\omega). \end{aligned}$$

Rearranging and applying Lemma 1.11 we have

$$\|(\{f+g\}|_{\mathbb{S}})_+\|_1 + \|(f|_{\mathbb{S}})_-\|_1 + \|(g|_{\mathbb{S}})_-\|_1 = \|(f|_{\mathbb{S}})_+\|_1 + \|(g|_{\mathbb{S}})_+\|_1 + \|(\{f+g\}|_{\mathbb{S}})_-\|_1.$$

By Lemma 1.8 and Definition 1.15 we have

$$\begin{aligned} \sum_{\omega \in \mathbb{S}} (f+g)(\omega) &= \|(\{f+g\}|_{\mathbb{S}})_+\|_1 - \|(\{f+g\}|_{\mathbb{S}})_-\|_1 \\ &= \|(f|_{\mathbb{S}})_+\|_1 - \|(f|_{\mathbb{S}})_-\|_1 + \|(g|_{\mathbb{S}})_+\|_1 - \|(g|_{\mathbb{S}})_-\|_1 \\ &= \sum_{\omega \in \mathbb{S}} f(\omega) + \sum_{\omega \in \mathbb{S}} g(\omega). \end{aligned}$$

Furthermore, given $\alpha \in \mathbb{R}$ choose $\tau \in \{-1, 1\}$ so that $\alpha = |\alpha|\tau$, so applying Lemma 1.8 and Definition 1.15 we deduce that

$$\begin{aligned} \sum_{\omega \in \mathbb{S}} (\alpha f)(\omega) &= \|(\alpha f|_{\mathbb{S}})_+\|_1 - \|(\alpha f|_{\mathbb{S}})_-\|_1 \\ &= |\alpha| \left\{ \|(\tau f|_{\mathbb{S}})_+\|_1 - \|(\tau f|_{\mathbb{S}})_-\|_1 \right\} \\ &= \tau |\alpha| \left\{ \|(f|_{\mathbb{S}})_+\|_1 - \|(f|_{\mathbb{S}})_-\|_1 \right\} = \alpha \sum_{\omega \in \mathbb{S}} f(\omega), \end{aligned}$$

as required. $\square$

Lemma 1.19. Suppose $f : \Omega \rightarrow \mathbb{R}$ is a function and subsets $\mathbb{A} \subseteq \mathbb{B} \subseteq \Omega$ such that $f|_{\mathbb{B}} \in \ell_+^1(\mathbb{B})$. Then $f|_{\mathbb{A}} \in \ell_+^1(\mathbb{A})$ and

$$\|f|_{\mathbb{A}}\|_1 = \sum_{\omega \in \mathbb{A}} f(\omega) \leq \sum_{\omega \in \mathbb{B}} f(\omega) = \|f|_{\mathbb{B}}\|_1.$$

Proof of Lemma 1.19. Since $f|_{\mathbb{B}} \in \ell_+^1(\mathbb{B})$ we have $(f|_{\mathbb{B}})_-(\omega) = 0$ for all $\omega \in \mathbb{B}$, and hence $(f|_{\mathbb{A}})_-(\omega) = (f|_{\mathbb{B}})_-(\omega) = 0$ for all $\omega \in \mathbb{A} \subseteq \mathbb{B}$. Note also that if $\mathbb{F} \subseteq \mathbb{A}$ then we also have $\mathbb{F} \subseteq \mathbb{B}$ and $(f|_{\mathbb{B}})_+(\omega) = (f|_{\mathbb{A}})_+(\omega)$ for all $\omega \in \mathbb{A}$, so that $\|(f|_{\mathbb{A}})_+\|_1 \leq \|(f|_{\mathbb{B}})_+\|_1$. Hence, we have

$$\begin{aligned} \sum_{\omega \in \mathbb{B}} f(\omega) - \sum_{\omega \in \mathbb{A}} f(\omega) &= \{ \|(f|_{\mathbb{B}})_+\|_1 - \|(f|_{\mathbb{B}})_-\|_1 \} - \{ \|(f|_{\mathbb{A}})_+\|_1 - \|(f|_{\mathbb{A}})_-\|_1 \} \\ &= \|(f|_{\mathbb{B}})_+\|_1 - \|(f|_{\mathbb{A}})_+\|_1 \geq 0, \end{aligned}$$

as required. Hence, we also have

$$\|f|_{\mathbb{A}}\|_1 = \sum_{\omega \in \mathbb{A}} |f(\omega)| = \sum_{\omega \in \mathbb{A}} f(\omega) \leq \sum_{\omega \in \mathbb{B}} f(\omega) = \sum_{\omega \in \mathbb{B}} |f(\omega)| = \|f|_{\mathbb{B}}\|_1 < \infty,$$

so $f|_{\mathbb{A}} \in \ell_+^1(\mathbb{A})$. $\square$

Lemma 1.20. Suppose $f : \Omega \rightarrow \overline{\mathbb{R}}$ is a function, and $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is a collection of pairwise disjoint sets with union $\mathbb{S} = \bigcup \mathcal{A}$ such that $f|_{\mathbb{S}} \in \ell^1(\mathbb{S})$. Then, there exists a function $\Phi \in \ell^1(\mathcal{A})$, defined by $\Phi(B) := \sum_{\omega \in B} f(\omega)$ for all $B \in \mathcal{A}$, such that

$$\sum_{\omega \in \mathbb{S}} f(\omega) = \sum_{B \in \mathcal{A}} \Phi(B) = \sum_{B \in \mathcal{A}} \left(\sum_{\omega \in B} f(\omega) \right). \quad (1.8)$$

Proof of Lemma 1.20. By Lemma 1.14 we have $\|f|_{\mathbb{S}}\|_1 = \|(f|_{\mathbb{S}})_-\|_1 + \|(f|_{\mathbb{S}})_+\|_1$ so both $\|(f|_{\mathbb{S}})_-\|_1$ and $\|(f|_{\mathbb{S}})_+\|_1$ are finite. For each $\tau \in \{-, +\}$ let's define $G^\tau : \mathcal{A} \rightarrow [0, \infty]$ by $G^\tau(B) := \|f_\tau|_B\|_1 = \|(f|_B)_\tau\|_1$ for all $B \in \mathcal{A}$. Since each $B \in \mathcal{A}$ we have $B \subseteq \bigcup \mathcal{A} = \mathbb{S}$ and so $G^\tau(B) = \|f_\tau|_B\|_1 = \|(f|_{\mathbb{S}})_\tau|_B\|_1$. Note also that since $\mathcal{A}$ is a collection of pairwise disjoint sets and $\mathbb{S} = \bigcup \mathcal{A}$, it follows that $\mathcal{A}$ is a partition $\mathbb{S}$. Hence, by Lemma 1.13 $\|G^\tau\|_1 = \|(f|_{\mathbb{S}})_\tau\|_1 < \infty$, so $G^\tau \in \ell^1_+(\mathcal{A})$ with $\text{Img}(G^\tau) \subseteq \mathbb{R}$ for each $\tau \in \{-, +\}$. By Definition 1.15 we have

$$\begin{aligned} \Phi(B) &= \sum_{\omega \in B} f(\omega) = \|(f|_B)_+\|_1 - \|(f|_B)_-\|_1 \\ &= G^+(B) - G^-(B) \in \mathbb{R}, \end{aligned}$$

for all $B \in \mathcal{A}$. Hence, by Lemma 1.8 we have $\Phi = G^+ - G^- \in \ell^1(\mathcal{A})$.

We define $\Phi_+ : \mathcal{A} \rightarrow \overline{\mathbb{R}}$ and $\Phi_- : \mathcal{A} \rightarrow \overline{\mathbb{R}}$ defined as in (1.7) so $\Phi_+(B) := \max\{\Phi(B), 0\}$ and $\Phi_-(B) = \max\{-\Phi(B), 0\}$ for all $B \in \mathcal{A}$. Hence, for all $B \in \mathcal{A}$ we have

$$\Phi_+(B) - \Phi_-(B) = \Phi(B) = G^+(B) - G^-(B).$$

Hence, we have $\Phi_+ + G^- = \Phi_- + G^+$ and $\Phi_-, \Phi_+, G^-, G^+ \in \ell^1_+(\mathcal{A})$. As such Lemma 1.11 yields $\|\Phi_+\|_1 + \|G^-\|_1 = \|\Phi_-\|_1 + \|G^+\|_1$. Hence, by applying Definition 1.15 to the sum over $\mathcal{A}$ we also have

$$\begin{aligned} \sum_{B \in \mathcal{A}} \Phi(B) &= \|\Phi_+\|_1 - \|\Phi_-\|_1 \\ &= \|G^+\|_1 - \|G^-\|_1 \\ &= \|(f|_{\mathbb{S}})_+\|_1 - \|(f|_{\mathbb{S}})_-\|_1 = \sum_{\omega \in \mathbb{S}} f(\omega), \end{aligned}$$

where we applied Definition 1.15 to the sum over $\mathbb{S}$ for the final equality. $\square$

Recall the concept of Cartesian products from (1.1).

Lemma 1.21. Suppose $\mathbb{J}$ is a non-empty finite index set and for each $i \in \mathbb{J}$ we have a set $\mathbb{A}_i$ and a function $h_i : \mathbb{A}_i \rightarrow \mathbb{R}$ such that $h_i \in \ell^1(\mathbb{A}_i)$. Define $H_{\mathbb{J}} : \prod_{i \in \mathbb{J}} \mathbb{A}_i \rightarrow \mathbb{R}$ by

$$H_{\mathbb{J}}(a) := \prod_{i \in \mathbb{J}} h_i(a_i),$$

for all $a = (a_i)_{i \in \mathbb{J}}$. Then, we have $H_{\mathbb{J}} \in \ell^1(\prod_{i \in \mathbb{J}} \mathbb{A}_i)$ with $\|H_{\mathbb{J}}\|_1 = \prod_{i \in \mathbb{J}} \|h_i\|_1$ and

$$\sum_{(a_i)_{i \in \mathbb{J}} \in \prod_{i \in \mathbb{J}} \mathbb{A}_i} H_{\mathbb{J}}(a) = \prod_{i \in \mathbb{J}} \left\{ \sum_{a_i \in \mathbb{A}_i} h_i(a_i) \right\}.$$

We shall prove [Lemma 1.21](#) for the special case $\mathbb{J} = \{0, 1\}$. The more general statement may be deduced from this special case through an inductive argument.

Proof of Lemma 1.21 with $\mathbb{J} = \{0, 1\}$. Let's begin by assuming that h_0 and h_1 are both non-negative so $h_0 \in \ell^1_+(\mathbb{A}_0)$ and $h_1 \in \ell^1_+(\mathbb{A}_1)$ and take $\varepsilon > 0$. Given any finite subset $\mathbb{F} \subseteq \mathbb{A}_0 \times \mathbb{A}_1$ we may choose finite subsets $\mathbb{F}_0 \subseteq \mathbb{A}_0$ and $\mathbb{F}_1 \subseteq \mathbb{A}_1$ and $\mathbb{F} \subseteq \mathbb{F}_0 \times \mathbb{F}_1 \subseteq \mathbb{A}_0 \times \mathbb{A}_1$, and also $\sum_{a_i \in \mathbb{F}_i} h_i(a_i) \geq \sum_{a_i \in \mathbb{A}_i} h_i(a_i) - \varepsilon = \|h_i\|_1 - \varepsilon$ for $i \in \{0, 1\}$. By the non-negativity of h_0, h_1 and hence $H_{\{0,1\}}$ we have

$$\begin{aligned} \sum_{(a_0, a_1) \in \mathbb{F}} H_{\{0,1\}}(a_0, a_1) &\leq \sum_{(a_0, a_1) \in \mathbb{F}_0 \times \mathbb{F}_1} H_{\{0,1\}}(a_0, a_1) \\ &= \sum_{(a_0, a_1) \in \mathbb{F}_0 \times \mathbb{F}_1} h_0(a_0) h_1(a_1) \\ &= \left(\sum_{a_0 \in \mathbb{F}_0} h_0(a_0) \right) \left(\sum_{a_1 \in \mathbb{F}_1} h_1(a_1) \right) \leq \|h_0\|_1 \|h_1\|_1. \end{aligned}$$

Since this holds for all finite subsets $\mathbb{F} \subseteq \mathbb{A}_0 \times \mathbb{A}_1$ we deduce that $\|H_{\{0,1\}}\|_1 \leq \|h_0\|_1 \|h_1\|_1$. On the other hand, since $\mathbb{F}_0 \times \mathbb{F}_1 \subseteq \mathbb{A}_0 \times \mathbb{A}_1$ is a finite set we also have

$$\begin{aligned} \|H_{\{0,1\}}\|_1 &\geq \sum_{(a_0, a_1) \in \mathbb{F}_0 \times \mathbb{F}_1} H_{\{0,1\}}(a_0, a_1) \\ &= \left(\sum_{a_0 \in \mathbb{F}_0} h_0(a_0) \right) \left(\sum_{a_1 \in \mathbb{F}_1} h_1(a_1) \right) \geq \max\{0, \|h_0\|_1 - \varepsilon\} \cdot \max\{0, \|h_1\|_1 - \varepsilon\}. \end{aligned}$$

Letting $\varepsilon \searrow 0$ shows that for non-negative functions we have

$$\|H_{\{0,1\}}\|_1 = \sum_{(a_0, a_1) \in \mathbb{A}_0 \times \mathbb{A}_1} H_{\{0,1\}}(a_0, a_1) = \left\{ \sum_{a_0 \in \mathbb{A}_0} h_0(a_0) \right\} \left\{ \sum_{a_1 \in \mathbb{A}_1} h_1(a_1) \right\} = \|h_0\|_1 \|h_1\|_1.$$

Now let's turn to the more general case where we assume $h_i \in \ell^1(\mathbb{A}_i)$ for all $i \in \{0, 1\}$. The result $\|H_{\{0,1\}}\|_1 = \|h_0\|_1 \|h_1\|_1$ follows from the non-negative case by [Lemma 1.6](#) since $|h_i| \in \ell^1_+(\mathbb{A}_i)$ for all $i \in \{0, 1\}$. For $i \in \{0, 1\}$ we decompose $h_i = h_i^+ - h_i^-$ as in [\(1.7\)](#) with $h_i^+, h_i^- \in \ell^1_+(\mathbb{A}_i)$ (for brevity, we write h_i^τ here, rather than $(h_i)_\tau$, where $\tau \in \{-, +\}$). Observe that for all $(a_0, a_1) \in \mathbb{A}_0 \times \mathbb{A}_1$ we have

$$\begin{aligned} H_{\{0,1\}}(a_0, a_1) &= \{h_0^+(a_0) - h_0^-(a_0)\} \{h_1^+(a_1) - h_1^-(a_1)\} \\ &= h_0^+(a_0)h_1^+(a_1) + h_0^-(a_0)h_1^-(a_1) - h_0^+(a_0)h_1^-(a_1) - h_0^-(a_0)h_1^+(a_1). \end{aligned} \tag{1.9}$$

Hence, for each $i \in \{0, 1\}$ we have $\sum_{a_i \in \mathbb{A}_i} h_i(a_i) = \|h_i^+\|_1 - \|h_i^-\|_1$ by Definition 1.15. Consequently, we have

$$\begin{aligned}
& \left\{ \sum_{a_0 \in \mathbb{A}_0} h_0(a_0) \right\} \left\{ \sum_{a_1 \in \mathbb{A}_1} h_1(a_1) \right\} \\
&= \left\{ \|h_0^+\|_1 - \|h_0^-\|_1 \right\} \left\{ \|h_1^+\|_1 - \|h_1^-\|_1 \right\} \\
&= \|h_0^+\|_1 \|h_1^+\|_1 + \|h_0^-\|_1 \|h_1^-\|_1 - \|h_0^+\|_1 \|h_1^-\|_1 - \|h_0^-\|_1 \|h_1^+\|_1 \\
&= \sum_{(a_0, a_1) \in \mathbb{A}_0 \times \mathbb{A}_1} h_0^+(a_0) h_1^+(a_1) + \sum_{(a_0, a_1) \in \mathbb{A}_0 \times \mathbb{A}_1} h_0^-(a_0) h_1^-(a_1) \\
&\quad - \sum_{(a_0, a_1) \in \mathbb{A}_0 \times \mathbb{A}_1} h_0^+(a_0) h_1^-(a_1) - \sum_{(a_0, a_1) \in \mathbb{A}_0 \times \mathbb{A}_1} h_0^-(a_0) h_1^+(a_1) \\
&= \sum_{(a_0, a_1) \in \mathbb{A}_0 \times \mathbb{A}_1} H_{\{0,1\}}(a_0, a_1),
\end{aligned}$$

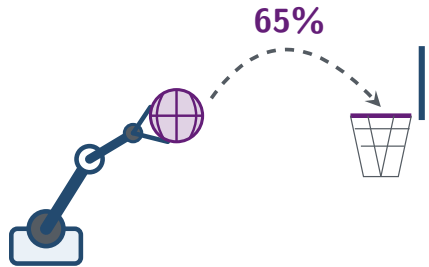
where the final equality follows from Lemma 1.18 combined with (1.9). $\square$

1.2 Foundational concepts in discrete probability

The modern mathematical foundations of probability were first formalised by Andrey Kolmogorov (Андрей Николаевич Колмогоров) [Kol50; SV06]. In this section we shall introduce this approach with an emphasis on discrete probability.

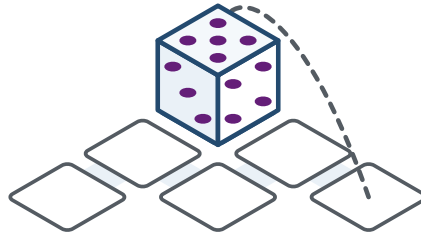
1.2.1 Stochastic scenarios

In mathematical probability we model wish to model stochastic scenarios or generating conditions. To set the scene, let's consider a few examples.



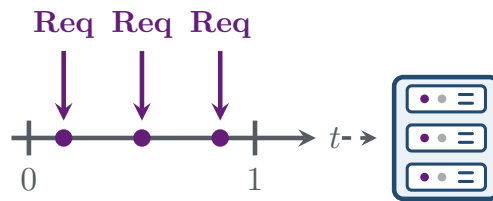
Example 1.22: A robot throws a basketball.

Example 1.22. A robot shoots a basketball with a with a 65% probability of success.



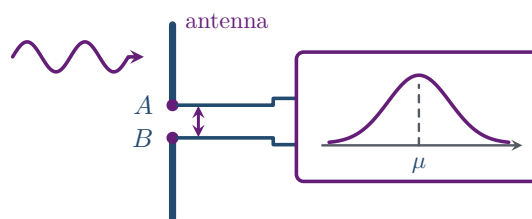
Example 1.23: A roll of the dice.

Example 1.23. We roll a dice whilst playing a board game.



Example 1.24: Requests to a web server.

Example 1.24. A web server has an expected number of three requests per second.



Example 1.25: Voltage across an antenna.

Example 1.25. The voltage across an antenna contains both signal and interference, due to background electromagnetic radiation and thermal fluctuations.

We will discuss these illustrative examples in more detail in the next few sections.

1.2.2 Possible outcomes and outcome spaces

A key component of the different stochastic scenarios encountered in Section 1.2.1 is that there were a variety mutually exclusive possible *outcomes*. In Example 1.22 the robot can either succeed or not succeed in getting the ball through the hoop; in Example 1.23 the dice may land with any one of its six sides facing up; in Example 1.24 the number of requests encountered within an hour could potentially be any one of a broad range of integers. Probability theory formalises this idea with the concept of an outcome space. Given stochastic scenario, the *outcome space* is the set of all possible outcomes.

Example 1.22 (continued) In our basketball playing robot scenario we have two possible outcomes $\Omega = \{H, M\}$, where “H” denotes the outcome that the event that the ball goes through the hoop and “M” denotes the outcome that the ball misses the hoop.

Example 1.23 (continued) Given a roll of the dice, the set of possible outcomes is $\Omega = \{1, \dots, 6\}$ where each number $a \in \{1, \dots, 6\}$ denotes the outcome that the a -th side of the dice facing up.

Example 1.24 (continued) We can model the web server over the course of an hour as a random environment with outcome space $\Omega = \mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$ where each number $k \in \mathbb{N}_0$ denotes the outcome in which the web server receives k requests.

Example 1.25 (continued) A natural model for the voltage encompasses a continuum of possible outcomes, so we take $\Omega = \mathbb{R}$.

1.2.3 Probability mass functions

A probability mass function assigns probabilities to individual outcomes.

Definition 1.26. Suppose Ω is an outcome space. A *probability mass function* on Ω is a map $\rho : \Omega \rightarrow [0, 1]$ such that $\sum_{\omega \in \Omega} \rho(\omega) = 1$.

For each of our first three examples (Examples 1.22 to 1.24) there is a natural probability mass function.

Example 1.22 (continued) For our basketball playing robot scenario we have $\Omega = \{H, M\}$ and we may define $\rho : \Omega \rightarrow [0, 1]$ by $\rho(H) := 0.65$ and $\rho(M) := 1 - \rho(H) = 0.35$.

Example 1.23 (continued) In our dice rolling example we have $\Omega = [6] = \{1, \dots, 6\}$ and we may define $\rho : \Omega \rightarrow [0, 1]$ by $\rho(\omega) := 1/6$ for all $\omega \in \Omega$.

Example 1.24 (continued) In our web server example we have $\Omega = \mathbb{N}_0$ and we may choose some $\lambda \in (0, \infty)$ and define $\rho_\lambda : \Omega \rightarrow [0, 1]$ by $\rho_\lambda(k) := \lambda^k e^{-\lambda} / k!$ for all $k \in \Omega$.

One should confirm that each of these examples (Examples 1.22 to 1.24) are probability mass functions (Definition 1.26), so that in each case the condition $\sum_{\omega \in \Omega} \rho(\omega) = 1$ is satisfied. In Example 1.24 the probability mass function ρ_λ involves a *parameter* which allows us to flexibly encode properties of a distribution without the necessity of directly specifying each value individually. If the expected number of requests per second is three (Example 1.24) then $\lambda = 3 \times 60^2$ is a reasonable choice, for reasons which will become clear later. The function ρ_λ given in Example 1.24 is referred to as a *Poisson* probability mass function in honour of the great mathematician and physicist Siméon Denis Poisson who investigated this model in the context of a statistical analysis of the number of wrongful convictions [GT12]. Later, we shall see connections between information theory and the estimation of parameters from data.

Example 1.25 (continued) In this scenario there is a natural model in which no countable set has probability one. In Lemma 1.27 we will see that such measures do not correspond to a probability mass function.

Given a set Ω , let's write $\Delta_{\text{pmf}}(\Omega)$ for the collection of all probability mass functions on Ω .

Lemma 1.27. Given any outcome space Ω we have

$$\Delta_{\text{pmf}}(\Omega) = \{ f \in \ell_+^1(\Omega) : \|f\|_1 = 1 \}. \quad (1.10)$$

Given $\rho \in \Delta_{\text{pmf}}(\Omega)$ the cozero set $\text{coz}(\rho) = \{ \omega \in \Omega : \rho(\omega) > 0 \}$ is countable.

Proof of Lemma 1.27. To establish (1.10), first suppose that $\rho \in \Delta_{\text{pmf}}(\Omega)$. Since $\text{Im}(\rho) \subseteq [0, 1]$ we have $\rho(\omega) \geq 0$ for all $\omega \in \Omega$. Hence, by Definition 1.15 we have

$$\|\rho\|_1 = \sum_{\omega \in \Omega} \rho(\omega) = 1,$$

so $\rho \in \ell_+^1(\Omega)$ and $\|\rho\|_1 = 1$.

Next, suppose that $\eta \in \ell_+^1(\Omega)$ and $\|\eta\|_1 = 1$. Then, for every $\omega \in \Omega$ we have

$$0 \leq \eta(\omega) = \sum_{a \in \{\omega\}} |\eta(a)| \leq \|\eta\|_1 = 1.$$

Consequently, $\eta : \Omega \rightarrow [0, 1]$ is a non-negative function and so

$$\sum_{\omega \in \Omega} \eta(\omega) = \|\eta\|_1 = 1,$$

so $\eta \in \Delta_{\text{pmf}}(\Omega)$. Hence, (1.10) holds. Finally, given $\rho \in \Delta_{\text{pmf}}(\Omega) \subseteq \ell_+^1(\Omega)$ we have $\text{coz}(\rho) = \{ \omega \in \Omega : \rho(\omega) > 0 \}$, which is countable by Lemma 1.7. $\square$

1.2.4 Discrete probability spaces

Recall that the power set $\mathcal{P}(\Omega)$ of Ω denotes the collection of all subsets $A \subseteq \Omega$.

Lemma 1.28. Suppose $\rho \in \Delta_{\text{pmf}}(\Omega)$. Then, for all $A \subseteq \Omega$ we have

$$\sum_{\omega \in A} \rho(\omega) = \|\rho|_A\|_1 = \|\rho \mathbf{1}_A\|_1 \in [0, 1].$$

Proof of Lemma 1.28. Suppose $\rho \in \Delta_{\text{pmf}}(\Omega) \subseteq [0, 1]^\Omega$. Then, given any $A \subseteq \Omega$ we have,

$$0 \leq \sum_{\omega \in A} \rho(\omega) = \|\rho|_A\|_1 = \|\rho \mathbf{1}_A\|_1 \leq \|\rho\|_1 = 1,$$

by the non-negativity of ρ , Lemma 1.10, Lemma 1.19 and Lemma 1.27. □

Given $\rho \in \Delta_{\text{pmf}}(\Omega)$ we can define a function $\mathbb{P}_\rho : \mathcal{P}(\Omega) \rightarrow [0, 1]$ by

$$\mathbb{P}_\rho(A) := \sum_{\omega \in A} \rho(\omega), \tag{1.11}$$

for all $A \subseteq \Omega$. Lemma 1.28 ensures that $\mathbb{P}_\rho$ is well-defined. We refer to $\mathbb{P}_\rho$ as the probability measure associated with ρ . In the context, of discrete probability theory we view subsets $A \subseteq \Omega$ as *events* and use the notation $\mathcal{E} = \mathcal{P}(\Omega)$ for the set of all events, provided that the outcome space Ω is either clear from the context or unimportant to the discussion at hand.

Definition 1.29. A *discrete probability space* is a triple $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ including an outcome space Ω , the set of all events $\mathcal{E} = \mathcal{P}(\Omega)$, and a discrete probability measure $\mathbb{P}_\rho$ associated with a probability mass function $\rho \in \Delta_{\text{pmf}}(\Omega)$.

We refer to $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ as the discrete probability space associated with the probability mass function $\rho \in \Delta_{\text{pmf}}(\Omega)$. Recall from Section 1.1 that $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is said to be a collection of pairwise disjoint subsets if $A_0 \cap A_1 = \emptyset$ for any pair of sets $A_0 \in \mathcal{A}$ and $A_1 \in \mathcal{A}$ such that $A_0 \neq A_1$. Theorem 1.30 gives the fundamental properties of a (discrete) probability space.

Theorem 1.30. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space associated with a probability mass function $\rho \in \Delta_{\text{pmf}}(\Omega)$. The countable set $\text{coz}(\rho) \subseteq \Omega$ satisfies $\mathbb{P}_\rho\{\text{coz}(\rho)\} = \mathbb{P}_\rho(\Omega) = 1$. Further more, given any collection $\mathcal{A} \subseteq \mathcal{E}$ of pairwise disjoint subsets of Ω we have

$$\mathbb{P}_\rho\left(\bigcup \mathcal{A}\right) = \sum_{E \in \mathcal{A}} \mathbb{P}_\rho(E). \tag{1.12}$$

Proof of Theorem 1.30. By Definition 1.26 and (1.11) we have $\mathbb{P}_\rho(\Omega) = \sum_{\omega \in \Omega} \rho(\omega) = 1$. By Lemma 1.19 we have $\rho|_{\bigcup \mathcal{A}} \in \ell_+^1(\bigcup \mathcal{A}) \subseteq \ell^1(\bigcup \mathcal{A})$. Consequently, it follows from Lemma 1.20 that, given any collection $\mathcal{A} \subseteq \mathcal{E}$ of pairwise disjoint subsets of Ω , we have

$$\mathbb{P}_\rho\left(\bigcup \mathcal{A}\right) = \sum_{\omega \in \bigcup \mathcal{A}} \rho(\omega) = \sum_{B \in \mathcal{A}} \left(\sum_{\omega \in B} \rho(\omega) \right) = \sum_{B \in \mathcal{A}} \mathbb{P}_\rho(B).$$

Note also that

$$\mathbb{P}_\rho\{\Omega \setminus \text{coz}(\rho)\} = \|\rho|_{\Omega \setminus \text{coz}(\rho)}\|_1 = \sup \left\{ \sum_{\omega \in \mathbb{F}} |\rho(\omega)| : \mathbb{F} \subseteq \Omega \setminus \text{coz}(\rho) \text{ is finite} \right\} = 0,$$

so by considering the pairwise disjoint collection $\{\text{coz}(\rho), \Omega \setminus \text{coz}(\rho)\}$ we have

$$1 = \mathbb{P}_\rho(\Omega) = \mathbb{P}_\rho\{\text{coz}(\rho)\} + \mathbb{P}_\rho\{\Omega \setminus \text{coz}(\rho)\} = \mathbb{P}_\rho\{\text{coz}(\rho) \cup \Omega \setminus \text{coz}(\rho)\} = \mathbb{P}_\rho\{\text{coz}(\rho)\},$$

as required. $\square$

Corollary 1.31. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space. We have $\mathbb{P}_\rho(\emptyset) = 0$ and whenever $A \subseteq B \subseteq \Omega$ we have $0 \leq \mathbb{P}_\rho(A) \leq \mathbb{P}_\rho(B)$.

Proof of Corollary 1.31 . By applying Theorem 1.30, (1.12) to $\{\Omega, \emptyset\}$ we obtain

$$\mathbb{P}_\rho(\Omega) = \mathbb{P}_\rho(\bigcup \{\Omega, \emptyset\}) = \mathbb{P}_\rho(\emptyset) + \mathbb{P}_\rho(\Omega),$$

so that $\mathbb{P}_\rho(\emptyset) = 0$. In addition, given $A \subseteq B$ we may write $B = A \cup (B \setminus A)$. Hence, by applying Theorem 1.30, (1.12) to the collection of pairwise disjoint sets $\{A, B \setminus A\}$ yields

$$\mathbb{P}_\rho(B) = \mathbb{P}_\rho(\bigcup \{A, B \setminus A\}) = \mathbb{P}_\rho(A) + \mathbb{P}_\rho(B \setminus A) \geq \mathbb{P}_\rho(A),$$

as required. $\square$

The following result (Corollary 1.32) is often referred to as the union bound.

Corollary 1.32. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space and $\mathcal{A} \subseteq \mathcal{E}$ is a countable collection. Then

$$\mathbb{P}_\rho\left(\bigcup \mathcal{A}\right) \leq \sum_{E \in \mathcal{A}} \mathbb{P}_\rho(E).$$

Proof of Corollary 1.32 . Since $\mathcal{A}$ is countable there is a bijection $\varphi : \mathbb{I} \rightarrow \mathcal{A}$ for some initial segment $\mathbb{I} \subseteq \mathbb{N}$. Let's define a sequence $(B_n)_{n \in \mathbb{N}}$ of sets $B_n \in \mathcal{E}$ as follows. For each $n \in \mathbb{N}$ let

$$B_n := \begin{cases} \varphi(n) \setminus \bigcup \{B_\ell : \ell < n\} & \text{if } n \in \mathbb{I} \\ \emptyset & \text{if } n \notin \mathbb{I}. \end{cases}$$

It follows that $\mathcal{B} = \{B_n : n \in \mathbb{N}\}$ is a pairwise disjoint collection with $\bigcup \mathcal{B} = \bigcup \mathcal{A}$. Hence, by [Theorem 1.30 \(1.12\)](#) combined with [Corollary 1.31](#) we have

$$\mathbb{P}_\rho\left(\bigcup \mathcal{A}\right) = \mathbb{P}_\rho\left(\bigcup \mathcal{B}\right) = \sum_{n \in \mathbb{N}} \mathbb{P}_\rho\{B_n\} \leq \sum_{n \in \mathbb{I}} \mathbb{P}_\rho\{\varphi(n)\} = \sum_{E \in \mathcal{A}} \mathbb{P}_\rho(E),$$

since $\varphi : \mathbb{I} \rightarrow \mathcal{A}$ is a bijection. $\square$

Lemma 1.33. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space and $A \subseteq \Omega$ has $\mathbb{P}_\rho(A) = 1$. Then, we have $\text{coz}(\rho) \subseteq A$.

Proof of Lemma 1.33. Suppose that $b \in \text{coz}(\rho) \setminus A$. Then by [\(1.11\)](#) we have

$$\mathbb{P}_\rho(\text{coz}(\rho) \setminus A) \geq \rho(b) > 0.$$

Hence, by [Theorem 1.30](#) applied to the collection of three pairwise disjoint sets $\{A, \text{coz}(\rho) \setminus A, \Omega \setminus (A \cup \text{coz}(\rho))\}$ we have

$$1 = \mathbb{P}_\rho(\Omega) = \mathbb{P}_\rho\{A\} + \mathbb{P}_\rho\{\text{coz}(\rho) \setminus A\} + \mathbb{P}_\rho\{\Omega \setminus (A \cup \text{coz}(\rho))\} > \mathbb{P}_\rho\{A\} = 1,$$

which would be a contradiction. Hence, we must have $\text{coz}(\rho) \setminus A = \emptyset$, so $\text{coz}(\rho) \subseteq A$. $\square$

1.2.5 Rings of sets and σ -additive functions

In this section we provide an interpretation of [Theorem 1.30](#), and also provide a converse result. First let's introduce a couple of useful concepts.

Definition 1.34. A *ring* of sets $\mathcal{F}$ is a collection of subsets $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ such that if $A \in \mathcal{F}$ and $B \in \mathcal{F}$ then $A \cup B \in \mathcal{F}$ and $A \setminus B \in \mathcal{F}$.

For example, the powerset $\mathcal{P}(\Omega)$ is a π -system.

Definition 1.35. A function $\mathbb{M} : \mathcal{F} \rightarrow [0, 1]$ defined on some collection of subsets $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ is said to be *σ -additive* if for any countable collection $\mathcal{A} \subseteq \mathcal{F}$ of pairwise disjoint subsets with $\bigcup \mathcal{A} \in \mathcal{F}$, we have

$$\mathbb{M}\left(\bigcup \mathcal{A}\right) = \sum_{E \in \mathcal{A}} \mathbb{M}(E). \quad (1.13)$$

Corollary 1.36. Given $\rho \in \Delta_{\text{pmf}}(\Omega)$ the function $\mathbb{P}_\rho : \mathcal{P}(\Omega) \rightarrow [0, 1]$ is σ -additive.

Proof of Corollary 1.36. Follows from [Theorem 1.30](#) and [Definition 1.35](#). $\square$

Hence, by [Corollary 1.36](#) the map $\mathbb{P}_\rho : \mathcal{P}(\Omega) \rightarrow [0, 1]$ is σ -additive function on a ring of sets $\mathcal{P}(\Omega)$ with $\mathbb{P}_\rho(\mathbb{S}) = 1$ for the countable set $\mathbb{S} = \text{coz}(\rho)$. It turns out that every such function is the probability measure for some probability mass function.

Proposition 1.37. Suppose Ω is a set, $\mathbb{S} \subseteq \Omega$ is a countable subset, and $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ is ring of sets $\{\mathbb{S}\} \cup \{\{\omega\} : \omega \in \mathbb{S}\} \subseteq \mathcal{F}$ and $\mathbb{M} : \mathcal{F} \rightarrow [0, 1]$ is a σ -additive function such that $\mathbb{M}(\mathbb{S}) = 1$. Then, there exists a probability mass function $\rho : \Omega \rightarrow [0, 1]$ such that for all $A \in \mathcal{F}$ we have $\mathbb{M}(A) = \sum_{\omega \in A} \rho(\omega) = \mathbb{P}_\rho(A)$.

Proof of Proposition 1.37. Consider an arbitrary set $B \in \mathcal{F}$. Since $\mathbb{S} \in \mathcal{F}$ and $\mathcal{F}$ is a ring we have $B \setminus \mathbb{S} \in \mathcal{F}$, so $\mathbb{S} \cup (B \setminus \mathbb{S}) \in \mathcal{F}$. Moreover, since $\mathcal{A}_0(B) := \{\mathbb{S}, B \setminus \mathbb{S}\}$ is a finite (and hence countable) partition of $\mathbb{S} \cup (B \setminus \mathbb{S}) \in \mathcal{F}$ it follows from (1.13) that

$$1 \geq \mathbb{M}\{\mathbb{S} \cup (B \setminus \mathbb{S})\} = \mathbb{M}(\mathbb{S}) + \mathbb{M}(B \setminus \mathbb{S}) = 1 + \mathbb{M}(B \setminus \mathbb{S}) \geq 1,$$

where we twice used the fact that $\text{Im}(\mathbb{M}) \subseteq [0, 1]$. Hence, $\mathbb{M}(B \setminus \mathbb{S}) = 0$ for all $B \in \mathcal{F}$. Let's define $\rho : \Omega \rightarrow [0, 1]$ defined by

$$\rho(\omega) := \begin{cases} \mathbb{M}(\{\omega\}) & \text{if } \omega \in \mathbb{S} \\ 0 & \text{if } \omega \notin \mathbb{S}. \end{cases}$$

Note that $\rho : \Omega \rightarrow [0, 1]$ is well defined since $\mathbb{M} : \mathcal{F} \rightarrow [0, 1]$ and $\{\{\omega\} : \omega \in \mathbb{S}\} \subseteq \mathcal{F}$. Next we let $\mathcal{A}_1(B) := \{\{\omega\} : \omega \in B \cap \mathbb{S}\} \cup \{B \setminus \mathbb{S}\} \subseteq \mathcal{F}$ so that $\mathcal{A}_1(B)$ is a countable collection of pairwise disjoint sets such that $B = \bigcup \mathcal{A}_1(B) = B$. Hence, by (1.13) we have

$$\begin{aligned} \mathbb{M}(B) &= \mathbb{M}\left\{\bigcup \mathcal{A}_1(B)\right\} = \sum_{E \in \mathcal{A}_1(B)} \mathbb{M}(E) \\ &= \mathbb{M}(B \setminus \mathbb{S}) + \sum_{\omega \in B \cap \mathbb{S}} \mathbb{M}(\{\omega\}) \\ &= 0 + \sum_{\omega \in B \cap \mathbb{S}} \rho(\omega) = \|\rho|_{B \setminus \mathbb{S}}\|_1 + \|\rho|_{B \cap \mathbb{S}}\|_1 \\ &= \|\rho|_B\|_1 = \sum_{\omega \in B} \rho(\omega), \end{aligned}$$

by [Lemma 1.13](#). In particular, by considering the case $\mathbb{S} \in \mathcal{F}$ and applying [Lemma 1.13](#) once more we have

$$1 = \mathbb{M}(\mathbb{S}) = \|\rho|_{\mathbb{S}}\|_1 + \|\rho|_{\Omega \setminus \mathbb{S}}\|_1 = \|\rho\|_1,$$

so that by [Lemma 1.27](#) we have $\rho \in \Delta_{\text{pmf}}(\Omega)$. □

1.2.6 Random variables on discrete probability spaces

A *random variable* on a discrete probability space $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is simply a function $X : \Omega \rightarrow \mathbb{R}$. Thus, in the context of a given discrete probability space $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$, the set of all random variables on $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is given by $\mathbb{R}^\Omega = \{f : \Omega \rightarrow \mathbb{R}\}$. We extend operations on $\mathbb{R}$ such as addition, scalar multiplication, products and so forth, through pointwise evaluation. For

example, if $X \in \mathbb{R}^\Omega$, $Y \in \mathbb{R}^\Omega$ are random variables and $\lambda \in \mathbb{R}$ is a scalar we define random variables $XY \in \mathbb{R}^\Omega$ and $\lambda X + Y \in \mathbb{R}^\Omega$ by

$$(XY)(\omega) = X(\omega)Y(\omega) \quad \text{and} \quad (\lambda X + Y)(\omega) = \lambda X(\omega) + Y(\omega),$$

for all $\omega \in \Omega$. Note that the probability mass function $\rho : \Omega \rightarrow [0, 1] \subseteq \mathbb{R}$ is itself a member of $[0, 1]^\Omega \subseteq \mathbb{R}^\Omega$, so we may view ρ as a random variable on the associated $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$. Given $X \in \mathbb{R}^\Omega$ we also have $X\rho \in \mathbb{R}^\Omega$.

Definition 1.38. The space of *integrable random variables* is

$$\mathcal{L}^1(\Omega, \mathbb{P}_\rho) := \{X \in \mathbb{R}^\Omega : X\rho \in \ell^1(\Omega)\} = \{X \in \mathbb{R}^\Omega : \|X\rho\|_1 < \infty\}.$$

Given $X \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$, its *expectation* is $\mathbb{E}_\rho(X) := \sum_{\omega \in \Omega} X(\omega)\rho(\omega)$.

Note that since if $\rho_0 : \Omega \rightarrow [0, 1]$ and $\rho_1 : \Omega \rightarrow [0, 1]$ are probability mass functions with $\mathbb{P}_{\rho_0} = \mathbb{P}_{\rho_1}$ we must have $\rho_0 = \rho_1$, so we may view $\mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ as depending on $\mathbb{P}_\rho$, as the notation suggests.

Lemma 1.39. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space and $A \subseteq \Omega$. Then $\mathbf{1}_A \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ and $\mathbb{E}_\rho(\mathbf{1}_A) = \mathbb{P}_\rho(A)$.

Proof of Lemma 1.39. By Lemmas 1.10, 1.19 and 1.27 and (1.11),

$$\|\mathbf{1}_A\rho\|_1 = \|\rho|_A\|_1 = \sum_{\omega \in A} \rho(\omega) = \mathbb{P}_\rho(A) \leq 1,$$

so $\mathbf{1}_A \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$. Moreover, $\mathbb{E}_\rho(\mathbf{1}_A) = \sum_{\omega \in A} \mathbf{1}_A(\omega)\rho(\omega) = \sum_{\omega \in A} \rho(\omega) = \mathbb{P}_\rho(A)$. $\square$

Lemma 1.40 confirms that $X \mapsto \mathbb{E}_\rho(X)$ is a linear functional of $\mathcal{L}^1(\Omega, \mathbb{P}_\rho)$.

Lemma 1.40. Suppose $X, Y \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ and $\lambda \in \mathbb{R}$. Then $\lambda X + Y \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ and

$$\mathbb{E}_\rho(\lambda X + Y) = \lambda \mathbb{E}_\rho(X) + \mathbb{E}_\rho(Y).$$

Proof of Lemma 1.40. Apply Lemmas 1.8 and 1.18 to the functions $X\rho$ and $Y\rho$. $\square$

Given random variables $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$ on a common outcome space Ω we let

$$\{X \leq Y\} := \{\omega \in \Omega : X(\omega) \leq Y(\omega)\}.$$

We will discuss this convention in more detail in Section 1.2.7.

Lemma 1.41. Suppose $X, Y \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ and $\mathbb{P}_\rho(X \leq Y) = 1$. Then $\mathbb{E}_\rho(X) \leq \mathbb{E}_\rho(Y)$.

Proof of Lemma 1.41. By Lemma 1.33, we have $\text{coz}(\rho) \subseteq \{X \leq Y\}$. Hence $X(\omega)\rho(\omega) \leq Y(\omega)\rho(\omega)$ for all $\omega \in \Omega$, so the conclusion of the lemma follows from Lemma 1.17. $\square$

1.2.7 Random elements and their distributions

A random variable assigns a real number to each outcome. It is often just as useful to assign other types object. For example, a word, a finite sequence, the state of a Markov chain, a graph or a matrix. This leads to the more general concept of random elements on discrete probability spaces.

Definition 1.42. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space and $\mathbb{A}$ is a set. A *random element taking values in $\mathbb{A}$* is a function $X : \Omega \rightarrow \mathbb{A}$. The set $\mathbb{A}$ is referred to as the *state space* of X , and its elements are called *states*.

Thus, a random variable is a random element taking values in $\mathbb{R}$.

Caution 1.43. There are two common conventions regarding the terminology of “random variables”. Some authors use the term “random variable” to mean what we refer to as a random element $X : \Omega \rightarrow \mathbb{A}$, taking values in a state space $\mathbb{A}$ which may differ from $\mathbb{R}$ [Le 22]. Random elements $Y : \Omega \rightarrow \mathbb{R}$ taking values in $\mathbb{R}$ are then referred to as “real-valued random variables”. For example, Le Gall [Le 22] uses the terms “random variable” and “real-valued random variable” in precisely this way. By contrast, we opt for the convention of Durrett [Dur19], whereby a “random variable” specifically refers to random elements taking values in $\mathbb{R}$, whilst “random element” is used for the more general concept (Definition 1.42).

Let’s now highlight some convenient notation. Given a random element $X : \Omega \rightarrow \mathbb{A}$, on a discrete probability space $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$, and a state $a \in \mathbb{A}$ we let

$$\{X = a\} := \{\omega \in \Omega : X(\omega) = a\},$$

and write $\mathbb{P}_\rho(X = a) = \mathbb{P}_\rho(\{\omega \in \Omega : X(\omega) = a\})$. More generally, suppose we have state spaces $\mathbb{A}_1, \dots, \mathbb{A}_m$, where $m \in \mathbb{N}$, along with a m -ary expression $\varphi(\cdot, \dots, \cdot)$, such that given any $(a_j)_{j \in [m]} \in \prod_{j \in [m]} \mathbb{A}_j$ the expression $\varphi(a_1, \dots, a_m)$ is an assertion regarding $a_1, \dots, a_m$. Then, given random elements $X_1, \dots, X_m$ be random elements on a common discrete probability space $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ with $X_j : \Omega \rightarrow \mathbb{A}_j$ for $j \in [m]$, we let

$$\{\varphi(X_1, \dots, X_m)\} := \{\omega \in \Omega : \varphi(X_1(\omega), \dots, X_m(\omega))\}.$$

For example, we let $\{X \in \mathbb{A}\}$ denote the set $\{\omega \in \Omega : X(\omega) \in \mathbb{A}\}$ and let $\{X \leq Y\}$ denote the set $\{\omega \in \Omega : X(\omega) \leq Y(\omega)\}$. Further more, given a random element $X : \Omega \rightarrow \mathbb{A}$ taking values in $\mathbb{A}$, along with a function $\varphi : \mathbb{A} \rightarrow \mathbb{B}$, we let $\varphi(X) : \Omega \rightarrow \mathbb{B}$ denote the function given by

$$\varphi(X)(\omega) = \varphi(X(\omega)),$$

for all $\omega \in \Omega$. Hence, $\varphi(X) : \Omega \rightarrow \mathbb{B}$ is a random element, taking values in the state space $\mathbb{B}$.

Definition 1.44. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space and $X : \Omega \rightarrow \mathbb{A}$ is a random element taking values in some state space $\mathbb{A}$. Then, the *distribution* of X is the function $\mathbb{P}_X : \mathcal{P}(\mathbb{A}) \rightarrow [0, 1]$ by $\mathbb{P}_X(S) := \mathbb{P}_\rho(\{X \in S\})$ for all $S \subseteq \mathbb{A}$. Further more, the *induced probability mass function* of X is the function $\rho_X : \mathbb{A} \rightarrow [0, 1]$ defined by $\rho_X(a) := \mathbb{P}_\rho(\{X = a\})$ for all $a \in \mathbb{A}$.

Lemma 1.45. Suppose $X : \Omega \rightarrow \mathbb{A}$ is a random element on a discrete probability space $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$. Then $\rho_X \in \Delta_{\text{pmf}}(\mathbb{A})$. Further more, given any $S \subseteq \mathbb{A}$, we have

$$\mathbb{P}_X(S) = \mathbb{P}_\rho(\{X \in S\}) = \sum_{a \in S} \rho_X(a). \quad (1.14)$$

Hence, $\mathbb{P}_X$ is the discrete probability measure associated with ρ_X .

Proof of Lemma 1.45. Let $\mathcal{J} = \{\{X = a\} : a \in S\}$ and notice that $\{X = a_0\} \cap \{X = a_1\} = \emptyset$ whenever $a_0 \neq a_1$, so $\mathcal{J}$ is a collection of pairwise disjoint sets, and also $\bigcup \mathcal{J} = \{X \in S\}$. Hence, by Definition 1.44 and Theorem 1.30 we have

$$\mathbb{P}_X(S) = \mathbb{P}_\rho(\{X \in S\}) = \sum_{a \in S} \mathbb{P}_\rho(\{X = a\}) = \sum_{a \in S} \rho_X(a).$$

Note also that $\{X \in \mathbb{A}\} = \{\omega \in \Omega : X(\omega) \in \mathbb{A}\} = \Omega$, so

$$\sum_{a \in \mathbb{A}} \rho_X(a) = \mathbb{P}_\rho(\{X \in \mathbb{A}\}) = \mathbb{P}_\rho(\Omega) = 1.$$

Note also that $\rho_X(a) = \mathbb{P}_\rho(\{X = a\}) \in [0, 1]$ by (1.11) and Lemma 1.28. Hence $\rho_X \in \Delta_{\text{pmf}}(\mathbb{A})$ by Definition 1.26 and (1.14) holds. $\square$

By Lemma 1.27, the cozero set of ρ_X is countable, even when $\mathbb{A}$ is uncountable.

Lemma 1.46. Suppose $X : \Omega \rightarrow \mathbb{A}$ is a random element on a discrete probability space $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ and $f : \mathbb{A} \rightarrow \mathbb{R}$. Then $\|f(X)\rho\|_1 = \|f\rho_X\|_1$. Moreover, the following conditions

$$(i) f(X) \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho), \quad (ii) f(X)\rho \in \ell^1(\Omega), \quad (iii) f\rho_X \in \ell^1(\mathbb{A}), \quad (iv) f \in \mathcal{L}^1(\mathbb{A}, \mathbb{P}_X),$$

are all equivalent. Further more, when $f(X) \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ we have

$$\mathbb{E}_\rho\{f(X)\} = \sum_{a \in \mathbb{A}} f(a) \rho_X(a) = \mathbb{E}_{\rho_X}(f). \quad (1.15)$$

Lemma 1.46 (1.15) is often referred to as the law of the unconscious statistician.

Proof of Lemma 1.46. We apply Lemma 1.8 and Lemma 1.13 to the partition $\mathcal{A} = \{ \{X = a\} : a \in \mathbb{A} \}$ to obtain

$$\begin{aligned}
\|f(X)\rho\|_1 &= \sum_{\omega \in \Omega} |f\{X(\omega)\} \rho(\omega)| \\
&= \sum_{a \in \mathbb{A}} \sum_{\omega \in \{X=a\}} |f\{X(\omega)\} \rho(\omega)| \\
&= \sum_{a \in \mathbb{A}} \sum_{\omega \in \{X=a\}} |f(a) \rho(\omega)| \\
&= \sum_{a \in \mathbb{A}} |f(a)| \left(\sum_{\omega \in \{X=a\}} |\rho(\omega)| \right) \\
&= \sum_{a \in \mathbb{A}} |f(a)| \mathbb{P}_\rho(\{X = a\}) \\
&= \sum_{a \in \mathbb{A}} |f(a)| \rho_X(a) = \|f\rho_X\|_1.
\end{aligned}$$

Hence, by Definition 1.38 (i), (ii), (iii), (iv) above are all equivalent. Moreover, by Lemma 1.18 and Lemma 1.20 applied twice we have

$$\begin{aligned}
\mathbb{E}_\rho\{f(X)\} &= \sum_{\omega \in \Omega} f(X)(\omega) \rho(\omega) \\
&= \sum_{a \in \mathbb{A}} \left\{ \sum_{\omega \in \{X=a\}} f(X)(\omega) \rho(\omega) \right\} \\
&= \sum_{a \in \mathbb{A}} f(a) \left\{ \sum_{\omega \in \{X=a\}} \rho(\omega) \right\} \\
&= \sum_{a \in \mathbb{A}} f(a) \mathbb{P}_\rho(\{X = a\}) \\
&= \sum_{a \in \mathbb{A}} f(a) \rho_X(a) = \mathbb{E}_{\rho_X}(f),
\end{aligned}$$

as required. □

The concept of the *distribution* of a random element also allows us to introduce the concept of two or more random variables being identically distributed.

Definition 1.47. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space and $\mathbb{A}$ is a state space. A collection of random elements $X_1, \dots, X_n : \Omega \rightarrow \mathbb{A}$ are said to be *identically distributed* if their respective distributions $\mathbb{P}_{X_1}, \dots, \mathbb{P}_{X_n}$ coincide $\mathbb{P}_{X_1} = \dots = \mathbb{P}_{X_n}$.

We note a couple of equivalent conditions.

Lemma 1.48. suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space and $X_1, \dots, X_n : \Omega \rightarrow \mathbb{A}$ are random elements. Then the following are equivalent conditions:

- (i) The random variables $X_1, \dots, X_n$ are identically distributed ie. $\mathbb{P}_{X_1} = \dots = \mathbb{P}_{X_n}$;
- (ii) Their respective probability mass functions coincide ie. $\rho_{X_1} = \dots = \rho_{X_n}$;
- (iii) Given any function $f : \mathbb{A} \rightarrow \mathbb{R}$ such that $f(X_1), \dots, f(X_n) \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ we have

$$\mathbb{E}_\rho\{f(X_1)\} = \dots = \mathbb{E}_\rho\{f(X_n)\}.$$

Proof of Lemma 1.48. We prove (i) $\Leftrightarrow$ (ii) and (i) $\Leftrightarrow$ (iii) separately.

(ii) $\Rightarrow$ (i). By Lemma 1.45 we have $\mathbb{P}_{X_1} = \dots = \mathbb{P}_{X_n}$ whenever $\rho_{X_1} = \dots = \rho_{X_n}$.

(i) $\Rightarrow$ (ii). By applying Lemma 1.45 to singleton sets of the form $\mathbb{S} = \{a\}$ for $a \in \mathbb{A}$, we have $\rho_{X_1} = \dots = \rho_{X_n}$ whenever $\mathbb{P}_{X_1} = \dots = \mathbb{P}_{X_n}$.

(iii) $\Rightarrow$ (i) By applying Lemma 1.39 and considering indicator functions $f = \mathbf{1}_A$ we have $\mathbb{P}_{X_1} = \dots = \mathbb{P}_{X_n}$ whenever (iii) holds.

(i) $\Rightarrow$ (iii) By Lemma 1.46 each of the expectations $\mathbb{E}_\rho\{f(X_1)\}, \dots, \mathbb{E}_\rho\{f(X_n)\}$ are determined by the respective distributions of $X_1, \dots, X_n$. $\square$

Lemma 1.49. We have $X \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ if and only if $|X| \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$. Further more, if $X \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ then $\mathbb{E}_\rho(|X|) = \|X\rho\|_1$.

Proof of Lemma 1.49. Since ρ is non-negative we have $|X|\rho = |X\rho|$. Thus, by Lemma 1.6 we have $X\rho \in \ell^1(\Omega)$ if and only if $|X|\rho \in \ell^1(\Omega)$, so by Definition 1.38 we have $X \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ if and only if $|X| \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$. By the second part of Definition 1.38 we see that $\mathbb{E}_\rho(|X|) = \| |X|\rho \|_1 = \|X\rho\|_1$ whenever $X \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$. $\square$

Corollary 1.50. Suppose $X, Y : \Omega \rightarrow \mathbb{R}$ with $Y \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ and $\mathbb{P}_\rho(\{|X| \leq |Y|\}) = 1$. Then $X \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ with $\mathbb{E}_\rho(|X|) \leq \mathbb{E}_\rho(|Y|) < \infty$.

Proof of Corollary 1.50. By Lemma 1.49 we have $|Y| \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ and $\|Y\rho\|_1 < \infty$. Also, by Lemma 1.33, we have $\text{coz}(\rho) \subseteq \{|X| \leq |Y|\}$. Hence, we have $|X(\omega)|\rho(\omega) \leq |Y(\omega)|\rho(\omega)$ for all $\omega \in \Omega$, so that by Corollary 1.12 we have $\|X\rho\|_1 \leq \|Y\rho\|_1 < \infty$, so that $X \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$. By Lemma 1.41 we also have $\mathbb{E}_\rho(|X|) \leq \mathbb{E}_\rho(|Y|) < \infty$. $\square$

Corollary 1.51. Suppose $X, Y : \Omega \rightarrow \mathbb{A}$ are random elements with $\mathbb{P}_\rho(\{X = Y\}) = 1$. Suppose that $f(Y) \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ where $f : \mathbb{A} \rightarrow \mathbb{R}$. Then $f(X) \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ with $\mathbb{E}_\rho\{f(X)\} = \mathbb{E}_\rho\{f(Y)\}$.

Proof of Corollary 1.51. Observe that if $X(\omega) = Y(\omega)$, for some $\omega \in \Omega$, then $f(X(\omega)) = f(Y(\omega))$. Hence, we have $\{X = Y\} \subseteq \{f(X) = f(Y)\} \subseteq \{|f(X)| \leq |f(Y)|\}$, so that $\mathbb{P}_\rho(\{f(X) = f(Y)\}) = 1$, by Corollary 1.31. Thus, since $f(Y) \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ it follows from

Corollary 1.50 that $f(X) \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$. Consequently, as $\mathbb{P}_\rho(\{f(X) = f(Y)\}) = 1$, by two applications of Lemma 1.41 we obtain. $\square$

1.2.8 Joint distributions and independence on discrete probability spaces

The immense value of random variables and random elements only becomes clear when we consider the way in which multiple random elements on common probability space $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ may combine and interact with one another. Recall the concept of Cartesian products from (1.1).

Definition 1.52. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space, that $\mathbb{F}$ is a non-empty finite set and for each $i \in \mathbb{F}$ we have a random element $X_i : \Omega \rightarrow \mathbb{A}_i$ taking values in the state space $\mathbb{A}_i$. The *joint random element* $X_{\mathbb{F}} = (X_i)_{i \in \mathbb{F}}$ is the map $X_{\mathbb{F}} : \Omega \rightarrow \prod_{i \in \mathbb{F}} \mathbb{A}_i$ defined by

$$X_{\mathbb{F}}(\omega) := (X_i(\omega))_{i \in \mathbb{F}} \in \prod_{i \in \mathbb{F}} \mathbb{A}_i,$$

for each $\omega \in \Omega$. Hence, $X_{\mathbb{F}}$ is a random element taking values in $\prod_{i \in \mathbb{F}} \mathbb{A}_i$. We then refer to the distribution of $X_{\mathbb{F}}$ (Definition 1.44) $\mathbb{P}_{X_{\mathbb{F}}} : \mathcal{P}(\prod_{i \in \mathbb{F}} \mathbb{A}_i) \rightarrow [0, 1]$, as the *joint distribution* of $(X_i)_{i \in \mathbb{F}}$. We also refer to $\rho_{X_{\mathbb{F}}} : \prod_{i \in \mathbb{F}} \mathbb{A}_i \rightarrow [0, 1]$, the induced probability mass function of $X_{\mathbb{F}}$, as the *joint probability mass function* of $(X_i)_{i \in \mathbb{F}}$.

For the special case of Definition 1.52 in which $\mathbb{F} = [n]$ for some $n \in \mathbb{N}$, and $\mathbb{A}_1 = \dots = \mathbb{A}_n = \mathbb{R}$, we refer to the random element $X_{[n]} : \Omega \rightarrow \mathbb{R}^n$ as a *random vector* and often denote $X_{[n]} = (X_i)_{i \in [n]}$ by $(X_1, \dots, X_n)$.

The most tractable collections of random elements $\{X_i : i \in \mathbb{F}\}$ are those for which there is no “interaction” between the different random elements within the collection, so the joint probability mass function *factorises*.

Definition 1.53. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space and $\mathbb{F}$ is a non-empty finite set. In addition, for each $i \in \mathbb{F}$ we have a random element $X_i : \Omega \rightarrow \mathbb{A}_i$, with probability mass function $\rho_{X_i} : \mathbb{A}_i \rightarrow [0, 1]$. We say that the indexed collection of random elements $(X_i)_{i \in \mathbb{F}}$ is *independent* if the joint distribution $\mathbb{P}_{X_{\mathbb{F}}} : \mathcal{P}(\prod_{i \in \mathbb{F}} \mathbb{A}_i) \rightarrow [0, 1]$ satisfies

$$\mathbb{P}_{X_{\mathbb{F}}}(\mathbf{E}) = \prod_{i \in \mathbb{F}} \mathbb{P}_{X_i}(\mathbf{E}_i), \quad (1.16)$$

for all $\mathbf{E} = \prod_{i \in \mathbb{F}} \mathbf{E}_i$ with $\mathbf{E}_j \subseteq \mathbb{A}_j$ for each $j \in \mathbb{F}$.

Later we will provide an information theoretic perspective on the concept of independence. Next, we observe that the property of independence is inherited by subsets of random variables.

Lemma 1.54. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space and $\mathbb{F}$ is a non-empty finite set. In addition, for each $i \in \mathbb{F}$ we have a random element $X_i : \Omega \rightarrow \mathbb{A}_i$. Then, the collection of random variables $(X_i)_{i \in \mathbb{F}}$ is independent if and only if $(X_i)_{i \in \mathbb{J}}$ is independent for every $\mathbb{J} \in \mathcal{P}(\mathbb{F}) \setminus \{\emptyset\}$.

Proof of Lemma 1.54. First suppose that $(X_i)_{i \in \mathbb{F}}$ is independent and take a subset $\mathbb{J} \subseteq \mathbb{F}$ such that $\mathbb{J} \neq \emptyset$. Suppose we have a sequence $(E_j)_{j \in \mathbb{J}}$ with $E_i \subseteq \mathbb{A}_i$ for each $i \in \mathbb{J}$ and let $E = \prod_{j \in \mathbb{J}} E_j \subseteq \prod_{j \in \mathbb{J}} \mathbb{A}_j$. We define $(\tilde{E}_i)_{i \in \mathbb{F}}$ by

$$\tilde{E}_i := \begin{cases} E_i & \text{if } i \in \mathbb{J} \\ \mathbb{A}_i & \text{if } i \notin \mathbb{J}, \end{cases}$$

and let $\tilde{E} = \prod_{j \in \mathbb{F}} \tilde{E}_j \subseteq \prod_{j \in \mathbb{F}} \mathbb{A}_j$. In particular, for all $j \in \mathbb{F} \setminus \mathbb{J}$ we have $\mathbb{P}_\rho(\{X_j \in \tilde{E}_j\}) = \mathbb{P}_\rho(\{X_j \in \mathbb{A}_j\}) = 1$. Then, by Definition 1.44 we have

$$\begin{aligned} \mathbb{P}_{X_{\mathbb{J}}}(E) &= \mathbb{P}_\rho(\{X_{\mathbb{J}} \in E\}) \\ &= \mathbb{P}_\rho\left(\bigcap_{j \in \mathbb{J}} \{X_j \in E_j\}\right) = \mathbb{P}_\rho\left(\bigcap_{j \in \mathbb{F}} \{X_j \in \tilde{E}_j\}\right) \\ &= \mathbb{P}_\rho(\{X_{\mathbb{F}} \in \tilde{E}\}) = \prod_{j \in \mathbb{F}} \mathbb{P}_\rho(\{X_j \in \tilde{E}_j\}) \\ &= \left\{ \prod_{j \in \mathbb{J}} \mathbb{P}_\rho(\{X_j \in \tilde{E}_j\}) \right\} \left\{ \prod_{j \in \mathbb{F} \setminus \mathbb{J}} \mathbb{P}_\rho(\{X_j \in \tilde{E}_j\}) \right\} = \prod_{j \in \mathbb{J}} \mathbb{P}_\rho(\{X_j \in E_j\}). \end{aligned}$$

Hence, the collection $(X_j)_{j \in \mathbb{J}}$ is independent.

The converse is immediate since $\mathbb{F} \in \mathcal{P}(\mathbb{F}) \setminus \{\emptyset\}$. □

An important special case of Definition 1.53 concerns pairs of random variables. In particular, we say that the random variables $X_0 : \Omega \rightarrow \mathbb{A}_0$ and $X_1 : \Omega \rightarrow \mathbb{A}_1$ are independent if the ordered pair $(X_i)_{i \in \{0,1\}}$ is independent in the sense of Definition 1.53. We denote independence between two random variables X_0 and X_1 by $X_0 \perp\!\!\!\perp X_1$. Before continuing we emphasise that the concept of independence given by Definition 1.53 differs from the strictly weaker concept of *pairwise* independence.

Definition 1.55. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space and $(X_i)_{i \in \mathbb{F}}$ is a collection of random elements $X_i : \Omega \rightarrow \mathbb{A}_i$ with a finite and non-empty index set $\mathbb{F}$. We say that $(X_i)_{i \in \mathbb{F}}$ are *pairwise independent* if $X_j \perp\!\!\!\perp X_k$ for any $j, k \in \mathbb{F}$ with $j \neq k$.

Clearly, when $|\mathbb{F}| = 2$ the concepts of pairwise independence and independence coincide. However, in general these concepts differ.

Caution 1.56. Independence (Definition 1.53) is strictly stronger than pairwise independence (Definition 1.55). For example, let $\Omega = \{0, 1\}^2$, give each outcome probability $1/4$, and define random variables $X, Y, Z : \Omega \rightarrow \mathbb{R}$ by

$$X(u, v) := u, \quad Y(u, v) := v, \quad Z(u, v) := (u + v) \bmod 2,$$

for all $(u, v) \in \Omega$. One may show that (X, Y, Z) are pairwise independent (Definition 1.55) and yet not independent (Definition 1.53).

To emphasise the distinction with pairwise independence some authors use the term “mutual independence” rather than “independence” (Definition 1.53). However, we shall simply use the term *independence*.

The next lemma provides some equivalent conditions for the independence of a collection of random variables.

Proposition 1.57. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space, that $\mathbb{F}$ is a non-empty finite index set and for each $i \in \mathbb{F}$ we have a random element $X_i : \Omega \rightarrow \mathbb{A}_i$, with distribution $\mathbb{P}_{X_i} : \mathcal{P}(\mathbb{A}_i) \rightarrow [0, 1]$ and probability mass function $\rho_{X_i} : \mathbb{A}_i \rightarrow [0, 1]$. Then, the following three conditions are equivalent:

- (i) Given any $E = \prod_{i \in \mathbb{F}} E_i \subseteq \prod_{i \in \mathbb{F}} \mathbb{A}_i$ we have $\mathbb{P}_{X_\mathbb{F}}(E) = \prod_{i \in \mathbb{F}} \mathbb{P}_{X_i}(E_i)$;
- (ii) Given any $a = (a_i)_{i \in \mathbb{F}} \in \prod_{i \in \mathbb{F}} \mathbb{A}_i$ we have $\rho_{X_\mathbb{F}}(a) = \prod_{i \in \mathbb{F}} \rho_{X_i}(a_i)$;
- (iii) Given any collection of functions $(\varphi_i)_{i \in \mathbb{F}}$ with $\varphi_i : \mathbb{A}_i \rightarrow \mathbb{R}$ and $\varphi_i \in \mathcal{L}^1(\mathbb{A}_i, \mathbb{P}_{X_i})$ for each $i \in \mathbb{F}$, the function $\varphi : \prod_{j \in \mathbb{F}} \mathbb{A}_j \rightarrow \mathbb{R}$ by $\varphi(a) = \prod_{j \in \mathbb{F}} \varphi_j(a_j)$ for all $a = (a_j)_{j \in \mathbb{F}} \in \prod_{i \in \mathbb{F}} \mathbb{A}_i$ we have $\varphi \in \mathcal{L}^1(\prod_{i \in \mathbb{F}} \mathbb{A}_i, \mathbb{P}_{X_\mathbb{F}})$ and

$$\mathbb{E}_\rho\{\varphi(X_\mathbb{F})\} = \prod_{i \in \mathbb{F}} \mathbb{E}_\rho\{\varphi_i(X_i)\}.$$

Proof of Proposition 1.57. We prove (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

(i) $\Rightarrow$ (ii). Given $a = (a_i)_{i \in \mathbb{F}} \in \prod_{i \in \mathbb{F}} \mathbb{A}_i$ may apply the special case of (i) with $E_i = \{a_i\} \subseteq \mathbb{A}_i$ for $i \in \mathbb{F}$ to obtain (ii).

(ii) $\Rightarrow$ (iii). Suppose that (ii) holds. For each $i \in \mathbb{F}$ we let $h_i : \mathbb{A}_i \rightarrow \mathbb{R}$ be the function given by $h_i(a_i) := \rho_{X_i}(a_i) \varphi_i(a_i)$ so that by Lemma 1.46 combined with $\varphi_i \in \mathcal{L}^1(\mathbb{A}_i, \mathbb{P}_{X_i})$ ensures that $h_i \in \ell^1(\mathbb{A}_i)$. By Lemma 1.21 it follows that $H_\mathbb{F} \in \ell^1(\prod_{i \in \mathbb{F}} \mathbb{A}_i)$ where

$$H_\mathbb{F}(a) := \prod_{i \in \mathbb{F}} h_i(a_i) = \prod_{i \in \mathbb{F}} \{\rho_{X_i}(a_i) \varphi_i(a_i)\} = \prod_{i \in \mathbb{F}} \rho_{X_i}(a_i) \prod_{i \in \mathbb{F}} \varphi_i(a_i) = \rho_{X_\mathbb{F}}(a) \varphi(a),$$

for all $a = (a_i)_{i \in \mathbb{F}} \in \prod_{i \in \mathbb{F}} \mathbb{A}_i$. Moreover, by Lemma 1.21 combined with Lemma 1.46 we have

$$\begin{aligned} \mathbb{E}_\rho\{\varphi(X_\mathbb{F})\} &= \sum_{a \in \prod_{i \in \mathbb{F}} \mathbb{A}_i} \varphi(a) \rho_{X_\mathbb{F}}(a) = \sum_{a \in \prod_{i \in \mathbb{F}} \mathbb{A}_i} H_\mathbb{F}(a) \\ &= \prod_{i \in \mathbb{F}} \left\{ \sum_{a_i \in \mathbb{A}_i} h_i(a_i) \right\} = \prod_{i \in \mathbb{F}} \mathbb{E}_\rho\{\varphi_i(X_i)\}, \end{aligned}$$

as required.

(iii) $\Rightarrow$ (i). Given an indexed collection of events $(E_i)_{i \in \mathbb{F}}$ we may apply the special case of (iii) with $\varphi_i = \mathbf{1}_{E_i}$ for each $i \in \mathbb{F}$ to obtain (i). $\square$

Any one of (i), (ii) and (iii) from [Proposition 1.57](#) may be treated as the definition of independence. The equivalence presented by [Proposition 1.57](#) ensures that there is no tension between our [Definition 1.53](#) and any one of these equivalent characterisations.

Corollary 1.58. Suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space and $(X_i)_{i \in \mathbb{F}}$ is an independent collection of random elements indexed by a finite set $\mathbb{F} \neq \emptyset$. Suppose $\mathbb{J}, \mathbb{K} \in \mathcal{P}(\mathbb{F}) \setminus \{\emptyset\}$ with $\mathbb{J} \cap \mathbb{K} = \emptyset$. Then $X_{\mathbb{J}} \perp\!\!\!\perp X_{\mathbb{K}}$. Furthermore, if $f : \prod_{j \in \mathbb{J}} \mathbb{A}_j \rightarrow \mathbb{R}$ and $g : \prod_{k \in \mathbb{K}} \mathbb{A}_k \rightarrow \mathbb{R}$ are real-valued functions with $f(X_{\mathbb{J}}), g(X_{\mathbb{K}}) \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$, then

$$\mathbb{E}_\rho\{f(X_{\mathbb{J}})g(X_{\mathbb{K}})\} = \mathbb{E}_\rho\{f(X_{\mathbb{J}})\} \mathbb{E}_\rho\{g(X_{\mathbb{K}})\}. \quad (1.17)$$

Proof of Corollary 1.58. By [Lemma 1.54](#) each of the indexed collections $(X_i)_{i \in \mathbb{J}}$, $(X_i)_{i \in \mathbb{K}}$ and $(X_i)_{i \in \mathbb{J} \cup \mathbb{K}}$ are independent. By [Proposition 1.57](#) (i) $\Rightarrow$ (ii) for each $\mathbb{L} \in \{\mathbb{J}, \mathbb{K}, \mathbb{J} \cup \mathbb{K}\}$ we have

$$\rho_{X_{\mathbb{L}}}(a_{\mathbb{L}}) = \prod_{i \in \mathbb{L}} \rho_{X_i}(a_i),$$

for all $a_{\mathbb{L}} := (a_i)_{i \in \mathbb{L}} \in \prod_{i \in \mathbb{L}} \mathbb{A}_i$. Hence, given any $a_{\mathbb{J} \cup \mathbb{K}} := (a_i)_{i \in \mathbb{J} \cup \mathbb{K}} \in \prod_{i \in \mathbb{J} \cup \mathbb{K}} \mathbb{A}_i$ we have

$$\rho_{X_{\mathbb{J} \cup \mathbb{K}}}(a_{\mathbb{J} \cup \mathbb{K}}) = \rho_{X_{\mathbb{J}}}(a_{\mathbb{J}}) \rho_{X_{\mathbb{K}}}(a_{\mathbb{K}}).$$

Consequently, by applying [Proposition 1.57](#) (ii) $\Rightarrow$ (i) once more we deduce that $X_{\mathbb{J}} \perp\!\!\!\perp X_{\mathbb{K}}$. By applying [Proposition 1.57](#) (i) $\Rightarrow$ (iii) we deduce (1.17). $\square$

Typically the most well-behaved collections of random elements $X_1, \dots, X_n$ are both independent [Definition 1.53](#) and identically distributed [Definition 1.47](#). We shall see an example of this in [Section 1.3.2](#).

1.3 High probability bounds

In this section we discuss the topic of high-probability bounds in which we seek to bound the probability of “rare” events. We begin with a simple but pivotal result known as “Markov’s inequality”. The importance of the result was highlighted by great Markov, but appears to have been used earlier by Chebyshev [[Mar84](#); [Cas23](#); [Tch67](#)]. We should be careful to distinguish between Markov’s inequality in probability ([Theorem 1.59](#) below) from a central result in approximation theory often referred to as the “Markov brother’s inequality” which bounds the magnitude of a polynomials derivatives [[Mar90](#); [Sha04](#)].

1.3.1 Markov’s inequality

In this section we discuss Markov’s inequality which is an elementary high-probability bound. Despite its simplicity it plays a pivotal role in establishing a range of more powerful high-probability bounds that follow.

Theorem 1.59. Let's suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space and $X \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ is integrable with $\mathbb{P}_\rho(X \geq 0) = 1$. Then, for all $\zeta > 0$,

$$\mathbb{P}_\rho(X \geq \zeta) \leq \mathbb{E}_\rho(X)/\zeta.$$

Proof of Theorem 1.59. Let $X^+ \in \mathbb{R}^\Omega$ denote the random variable given by $X^+(\omega) = \max\{X(\omega), 0\}$ for all $\omega \in \Omega$, so that $\mathbb{P}_\rho(\{X^+ = X\}) = \mathbb{P}_\rho(\{X \geq 0\}) = 1$ and by Corollary 1.51 we have $X^+ \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$ with $\mathbb{E}_\rho(X^+) = \mathbb{E}_\rho(X)$. Furthermore, given any $\omega \in \Omega$ we have $X^+(\omega) \geq 0$ so that

$$X^+ = X^+ \mathbf{1}_{\{X^+ < 0\}} + X^+ \mathbf{1}_{\{X^+ \in [0, \zeta)\}} + X^+ \mathbf{1}_{\{X^+ \geq \zeta\}} = X \mathbf{1}_{\{X \in [0, \zeta)\}} + X \mathbf{1}_{\{X \geq \zeta\}} \geq \zeta \mathbf{1}_{\{X \geq \zeta\}},$$

Consequently, by lemmas 1.41, 1.40 and 1.39 we have

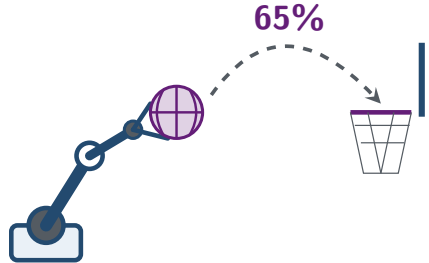
$$\mathbb{E}_\rho(X) = \mathbb{E}_\rho(X^+) \geq \mathbb{E}_\rho(\zeta \mathbf{1}_{\{X \geq \zeta\}}) = \zeta \mathbb{E}_\rho(\mathbf{1}_{\{X \geq \zeta\}}) = \zeta \mathbb{P}_\rho(X \geq \zeta).$$

Dividing through by $\zeta > 0$ completes the proof of the lemma. $\square$

A more comprehensive discussion Markov's inequality including improvements is [Hub19].

1.3.2 Bernoulli's law of large numbers

In this section we shall demonstrate the strength of Markov's inequality through an application to sums of independent and identically distributed Bernoulli random variables. First let's first review the concept of a Bernoulli random variable.



Example 1.22: A robot throws a basketball.

Example 1.22. A robot shoots a basketball with a with a 65% probability of success.

A Bernoulli random variable formalises the concept of a stochastic scenario or random experiment with only two possible outcomes. In this section we shall leave the probability space $(\Omega, \mathcal{E}, \mathbb{P})$ implicit and let $\mathbb{E}$ denote the corresponding expectation map (Definition 1.38).

Definition 1.60. A *Bernoulli* random variable X is a random variable $X : \Omega \rightarrow \mathbb{R}$ which takes values in $\{0, 1\}$. Given $p \in [0, 1]$ we write $X \sim \text{Bern}(p)$ to mean that X is a Bernoulli random variable with $\mathbb{E}(X) = p$.

Observe that if $X \sim \text{Bern}(p)$ then its distribution $\mathbb{P}_X : \mathcal{P}(\mathbb{R}) \longrightarrow [0, 1]$ has probability mass function $\rho_X : \mathbb{R} \rightarrow [0, 1]$ by

$$\rho_X(x) := \begin{cases} 1-p & \text{if } x = 0 \\ p & \text{if } x = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Hence, the family of distributions for Bernoulli random variables is parameterised by a single parameter $p \in [0, 1]$. We often refer $p = \mathbb{E}(X)$ as the “success probability”. By aggregating multiple Bernoulli random variables $X_1, \dots, X_n \sim \text{Bern}(p)$ we obtain a Binomial random variable.

Definition 1.61. A *binomial* random variable $Y : \Omega \rightarrow \mathbb{R}$ is a random sum of the form $Y = \sum_{i=1}^n X_i$ where $n \in \mathbb{N}$ and $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ are independent and identically distributed Bernoulli random variables on a common probability space $(\Omega, \mathcal{E}, \mathbb{P})$. Given $n \in \mathbb{N}$ and $p \in [0, 1]$ we write $Y \sim \text{Bin}(n, p)$ to mean that $Y = \sum_{i=1}^n X_i$ with n independent terms $X_i \sim \text{Bern}(p)$.

Given a collection of n independent and identically distributed random variables $X_1, \dots, X_n$ with a common success probability p , a natural estimator for p is the sample mean

$$\hat{p}_n := \frac{1}{n} \sum_{i=1}^n X_i.$$

Thus, $n\hat{p}_n \sim \text{Bin}(n, p)$. Bernoulli’s law of large numbers implies that, in a certain probabilistic sense, the random variable $\hat{p}_n$ converges to p .

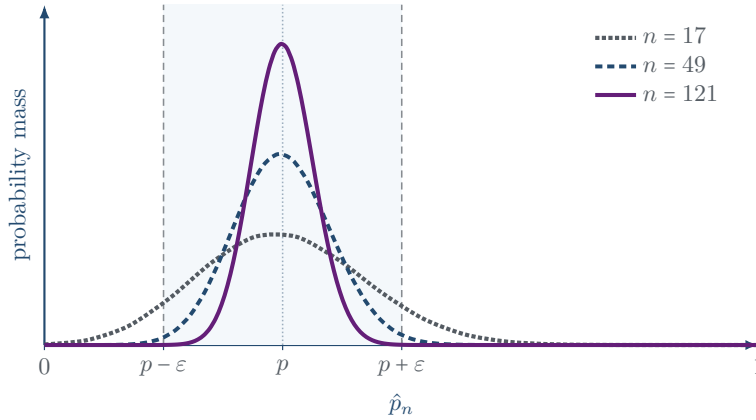


Figure 1.2. Distributions of the sample proportion for $p = \frac{1}{3}$ and $\varepsilon = \frac{1}{6}$. Each visible bar has area proportional to its probability. Red bars represent $|\hat{p}_n - p| \geq \varepsilon$; the tolerance band is fixed.

Theorem 1.62. Suppose $p \in [0, 1]$ and for each $n \geq 1$, we have a random variable $\hat{p}_n$ on a probability space $(\Omega_n, \mathcal{E}_n, \mathbb{P}_n)$ such that $n\hat{p}_n \sim \text{Bin}(n, p)$. Then, given any $\varepsilon > 0$, we have

$$\lim_{n \rightarrow \infty} \mathbb{P}_n(|\hat{p}_n - p| \geq \varepsilon) = 0.$$

Theorem 1.62 dates back to the pioneering work of Bernoulli [Ber13]. Bernoulli’s original proof involved some fairly involved combinatorial arguments. In the next section we will discuss how Markov’s bound (Theorem 1.59) can be “boosted” to provide a relatively straightforward proof.

1.3.3 The Chernoff–Chebyshev–Rubin method

... many things are only trivial once you know them

Chernoff [Bat96]

At first glance, Markov’s inequality (Theorem 1.59) appears quite limited. Indeed, to prove Theorem 1.62 we wish to show that as a sequence, the probability of the random variable $\hat{p}_n$ exceeding a given value $p + \varepsilon$ converges to zero as $n \nearrow \infty$; to directly apply Theorem 1.59 we would require that the expectations $\mathbb{E}_n(\hat{p}_n)$ converge to zero, which they don’t! Nonetheless, through a simple transformation we obtain a much more powerful version of the bound. First recall that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be *non-decreasing* if $f(x_0) \leq f(x_1)$ whenever $x_0 \leq x_1$.

Theorem 1.63. Let’s suppose $(\Omega, \mathcal{E}, \mathbb{P}_\rho)$ is a discrete probability space, $f : \mathbb{R} \rightarrow [0, \infty)$ be a non-decreasing function and X is a random variable such that $f(X) \in \mathcal{L}^1(\Omega, \mathbb{P}_\rho)$. Then, for all $\zeta \in \mathbb{R}$ with $f(\zeta) > 0$ we have

$$\mathbb{P}_\rho(X \geq \zeta) \leq \mathbb{E}_\rho\{f(X)\}/f(\zeta).$$

Proof of Theorem 1.63. Since f is non-decreasing $\{X \geq \zeta\} \subseteq \{f(X) \geq f(\zeta)\}$. Hence,

$$\mathbb{P}_\rho(X \geq \zeta) \leq \mathbb{P}_\rho\{f(X) \geq f(\zeta)\} \leq \mathbb{E}_\rho\{f(X)\}/f(\zeta),$$

where the second inequality follows from Theorem 1.59 applied with $f(X)$ and $f(\zeta)$ in place of X and ζ , respectively. $\square$

The method became popular after a celebrated paper of Chernoff [Che52]. Chernoff attributes his knowledge of the technique to Rubin [Bat96]. Chebyshev had used the approach in the nineteenth century [Tch67].

1.3.4 Hoeffding’s bound and the Kullback–Leibler divergence

Hoeffding [Hoe63] provided a general tail bound, a special case of which (Theorem 1.65) which implies Theorem 1.62. We can explain Hoeffding’s bound in terms of a special case of the Kullback–Leibler divergence.

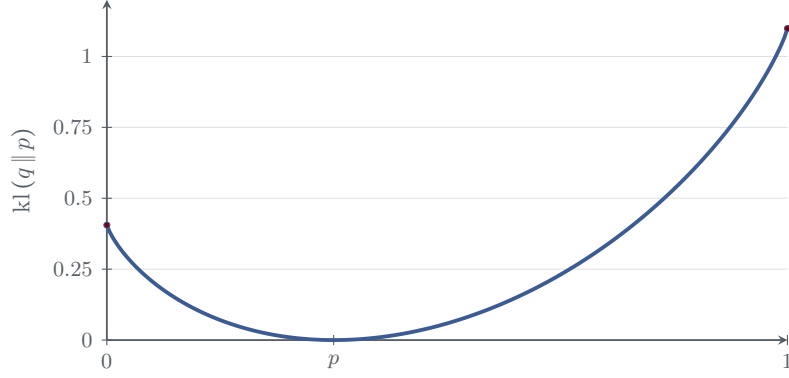


Figure 1.3. The binary Kullback–Leibler divergence $q \mapsto \text{kl}(q \parallel p)$.

Definition 1.64. Given $p \in (0, 1)$ and $q \in [0, 1]$, the *binary Kullback–Leibler divergence* of q with respect to p is

$$\text{kl}(q \parallel p) := q \log\left(\frac{q}{p}\right) + (1 - q) \log\left(\frac{1 - q}{1 - p}\right) \in [0, \infty),$$

where we use the convention $0 \log 0 := \lim_{x \searrow 0} x \log x = 0$ for $q \in \{0, 1\}$.

Let's also write $\underline{\text{kl}}(r \parallel p) := \text{kl}(\min\{\max\{r, 0\}, 1\} \parallel p)$ for $p \in (0, 1)$ and $r \in \mathbb{R}$.

Theorem 1.65. Suppose $p \in (0, 1)$ and for each $n \geq 1$, we have a random variable $\hat{p}_n$ on a probability space $(\Omega, \mathcal{E}, \mathbb{P})$ such that $n \hat{p}_n \sim \text{Bin}(n, p)$. Then, for $\varepsilon > 0$ we have

$$\mathbb{P}(\hat{p}_n \geq p + \varepsilon) \leq e^{-n \underline{\text{kl}}(p + \varepsilon \parallel p)} \quad \text{and} \quad \mathbb{P}(\hat{p}_n \leq p - \varepsilon) \leq e^{-n \underline{\text{kl}}(p - \varepsilon \parallel p)}. \quad (1.18)$$

Lemma 1.66. Suppose $p \in (0, 1)$ and for each $n \geq 1$, we have a random variable $\hat{p}_n$ on a probability space $(\Omega, \mathcal{E}, \mathbb{P})$ such that $n \hat{p}_n \sim \text{Bin}(n, p)$. Given $q \in (p, 1)$ and $\lambda > 0$,

$$\frac{1}{n} \log \mathbb{P}(\hat{p}_n \geq q) \leq \log\{p e^\lambda - p + 1\} - \lambda q. \quad (1.19)$$

Proof of Lemma 1.66. We apply the Chernoff–Chebyshev–Rubin method (Theorem 1.63), with the function $x \mapsto e^{n\lambda x}$ to obtain

$$\mathbb{P}(\hat{p}_n \geq q) = \mathbb{P}(e^{n\lambda \hat{p}_n} \geq e^{n\lambda q}) \leq e^{-n\lambda q} \mathbb{E}(e^{n\lambda \hat{p}_n}), \quad (1.20)$$

where we applied Theorem 1.59 for the inequality. Now we apply independence to evaluate the expectation as follows,

$$\begin{aligned} \mathbb{E}(e^{n\lambda \hat{p}_n}) &= \mathbb{E}(e^{\lambda \sum_{i=1}^n X_i}) = \mathbb{E}\left(\prod_{i=1}^n e^{\lambda X_i}\right) \\ &= \prod_{i=1}^n \mathbb{E}(e^{\lambda X_i}) = \{p e^\lambda - p + 1\}^n, \end{aligned}$$

where we applied [Proposition 1.57](#) for the penultimate equality and $X_i \sim \text{Bern}(p)$ for the final equality. By substituting our formula for $\mathbb{E}(e^{n\lambda\hat{p}_n})$ back into (1.20) and taking logarithms we obtain (1.19). $\square$

To complete the proof of [Theorem 1.65](#) we apply elementary calculus to optimise the bound.

Lemma 1.67. Given $p \in (0, 1)$ and $q \in (p, 1)$ let $\psi_{p,q} : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$\psi_{p,q}(\lambda) := \log(1 - p + pe^\lambda) - \lambda q,$$

for all $\lambda \in \mathbb{R}$. Then, we have

$$\inf\{\psi_{p,q}(\lambda) : \lambda \in (0, \infty)\} = -\text{kl}(q \| p).$$

Proof of Lemma 1.67. Let $\lambda_* := \log\{q(1-p)\} - \log\{p(1-q)\}$, which is strictly positive since $q > p$. By differentiating we see that $\psi_{p,q}$ strictly decreasing for $\lambda < \lambda_*$ and strictly increasing for $\lambda > \lambda_*$. Hence, $\psi_{p,q}$ attains its minimum at λ_* , and substituting in this value gives we obtain the minimum of $\psi_{p,q}(\lambda_*) = -\text{kl}(q \| p)$. $\square$

We can now complete the proof of [Theorem 1.65](#).

Proof of Theorem 1.65. Let's begin by proving the a bound on the upper tail. We shall show that for any $\varepsilon > 0$ we have

$$\mathbb{P}(\hat{p}_n \geq p + \varepsilon) \leq e^{-n\underline{\text{kl}}(p+\varepsilon \| p)}. \quad (1.21)$$

First suppose $\varepsilon \in (0, 1-p)$ and let $q = p + \varepsilon$. By combining lemmas [1.66](#) and [1.67](#) we have

$$\frac{1}{n} \log \mathbb{P}(\hat{p}_n \geq p + \varepsilon) \leq -\text{kl}(q \| p) = -\underline{\text{kl}}(p + \varepsilon \| p).$$

Exponentiating gives the upper bound in (1.21) provided $\varepsilon \in (0, 1-p)$. To establish the bound for $\varepsilon \geq 1-p$ we deduce from [Corollary 1.31](#) that for every $\zeta \in (0, 1-p)$ we have

$$\mathbb{P}(\hat{p}_n \geq p + \varepsilon) \leq \mathbb{P}(\hat{p}_n \geq 1 - \zeta) \leq e^{-n\underline{\text{kl}}(1-\zeta \| p)},$$

and letting $\zeta \searrow 0$ we deduce the upper bound in (1.21) for all $\varepsilon > 0$. By symmetry we also have the upper bound on the lower tail

$$\mathbb{P}(\hat{p}_n \leq p - \varepsilon) \leq e^{-n\underline{\text{kl}}(p-\varepsilon \| p)}, \quad (1.22)$$

as required. $\square$

Theorem 1.68. Suppose $p \in [0, 1]$ and for each $n \geq 1$, we have a random variable $\hat{p}_n$ on a probability space $(\Omega, \mathcal{E}, \mathbb{P})$ such that $n\hat{p}_n \sim \text{Bin}(n, p)$. Then, for $\varepsilon > 0$ we have

$$\mathbb{P}(|\hat{p}_n - p| \geq \varepsilon) \leq 2e^{-2n\varepsilon^2}. \quad (1.23)$$

We deduce [Theorem 1.68](#) from [Theorem 1.65](#) via the following lemma.

Lemma 1.69. Given any $p \in (0, 1)$ and $q \in [0, 1]$ we have $\text{kl}(q \parallel p) \geq 2(q - p)^2$.

Proof. Let's fix $p \in (0, 1)$ and define a function $g_p : [0, 1] \rightarrow \mathbb{R}$ by $g_p(q) := \text{kl}(q \parallel p) - 2(q - p)^2$ for all $q \in [0, 1]$. By computing derivatives we have $g_p(p) = g'_p(p) = 0$ and $g''_p(q) \geq 0$ for all $q \in (0, 1)$. Hence $g_p(q) \geq 0$ for all $q \in [0, 1]$, and adding $2(q - p)^2$ to both sides of the inequality gives the lemma. $\square$

Proof of Theorem 1.68. Let's begin by establishing the upper bound

$$\mathbb{P}(\hat{p}_n \geq p + \varepsilon) \leq \exp(-2n\varepsilon^2). \quad (1.24)$$

Let's consider three cases. If $p = 1$ then $\mathbb{P}\{\hat{p}_n \neq 1\} = 0$ which implies (1.24). Now suppose $p < 1$. If $\varepsilon > 1 - p$ then $\{\hat{p}_n \geq p + \varepsilon\} = \emptyset$, so (1.24) follows from $\mathbb{P}(\emptyset) = 0$ by [Corollary 1.31](#). Finally, let's suppose $p < 1$ and $\varepsilon \in (0, 1 - p)$. Then, by applying [Theorem 1.65](#) combined with [Lemma 1.69](#) we obtain (1.24). Hence, we conclude that (1.24) holds for all $p \in [0, 1]$ and $\varepsilon > 0$.

By symmetry we also have $\mathbb{P}(\hat{p}_n \leq p - \varepsilon) \leq \exp(-2n\varepsilon^2)$ for all $p \in [0, 1]$ and $\varepsilon > 0$. By applying the union bound ([Corollary 1.32](#)) we deduce the two sided bound (1.23). $\square$

Proof of Theorem 1.62. Apply the quantitative bound given by [Theorem 1.68](#). $\square$

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